Job Matching with Subsidy and Taxation

Fuhito Kojima, Ning Sun, and Ning Neil Yu

Online Appendix

OA.1 The Substitutes Condition and $M^\natural$-Concavity

This section briefly reviews the connection between the substitutes condition (which is central to economic theory) and $M^\natural$-concavity (which is the mainstay of discrete convex analysis), and relevant results used in this paper (see Murota, 1996, 2003, 2016, 2021b; Murota and Shioura, 1999; Fujishige and Yang, 2003; Milgrom and Strulovici, 2009; Shioura and Tamura, 2015, for more details).

Denote the **indicator function for** $P \in \mathcal{P}$ by $\mathbf{i}^P \in \mathbb{R}^{\mathcal{P}}$, that is, $\mathbf{i}^P(P) = 1$ while $\mathbf{i}^P(P') = 0$ for each $P' \neq P$. A vectorial function U is **$M^\natural$-concave** if for all $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $P \in \mathcal{P}$ such that $\mathbf{z}(P) > \mathbf{z}'(P)$, either $U(\mathbf{z}) + U(\mathbf{z}') \leq U(\mathbf{z} - \mathbf{i}^P) + U(\mathbf{z}' + \mathbf{i}^P)$, or there exists $P' \in \mathcal{P}$ such that $\mathbf{z}'(P') > \mathbf{z}(P')$ and $U(\mathbf{z}) + U(\mathbf{z}') \leq U(\mathbf{z} - \mathbf{i}^P + \mathbf{i}^{P'}) + U(\mathbf{z}' + \mathbf{i}^P - \mathbf{i}^{P'})$, or both.

There is an analogous concept for revenue functions. Given any set of workers $A \subset D$ and $d \in D$, we write $A + d \coloneqq A \cup \{d\}$ and $A - d \coloneqq A \setminus \{d\}$, and follow this convention for other sets too. A revenue function R is **discrete concave** if for all $A, B \subset D$ and $d \in A \setminus B$, either $R(A) + R(B) \leq R(A - d) + R(B + d)$, or there exists $d' \in B \setminus A$ such that $R(A) + R(B) \leq R(A - d + d') + R(B + d - d')$, or both.[1]

The following characterization of the substitutes condition by Fujishige and Yang (2003) brings together the economics literature on the substitutes condition and discrete convex analysis.

Lemma OA.1 (Fujishige and Yang (2003))**.** *A revenue function R satisfies the substitutes condition if and only if it is discrete concave.*

Given Lemma OA.1, it is straightforward to prove the following known equivalence between the vectorial substitutes condition and $M^\natural$-concavity (Shioura and Tamura, 2015, Theorem 4.1).

[1] A discrete concave revenue function is called an $M^\natural$-concave set function by Murota (2016). We rename it to avoid ambiguity.

Lemma OA.2. *A vectorial function satisfies the vectorial substitutes condition if and only if it is $M^\natural$-concave.*

Due to this lemma, we use the concept of $M^\natural$-concavity interchangeably with the vectorial substitutes condition, and are allowed to utilize powerful results in discrete convex analysis in our proofs. For example, the combination of the following two conditions characterize $M^\natural$-concavity and thus the vectorial substitutes condition.[2]

Theorem OA.1 (Murota and Shioura (2018))**.** *A vectorial function U is $M^\natural$-concave if and only if the following two conditions are both satisfied:*

- *for all $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ such that $|\mathbf{z}| > |\mathbf{z}'|$, there exists $P \in \mathcal{P}$ such that $\mathbf{z}(P) > \mathbf{z}'(P)$ and $U(\mathbf{z}) + U(\mathbf{z}') \leq U(\mathbf{z} - \mathbf{i}^P) + U(\mathbf{z}' + \mathbf{i}^P)$;*
- *for all $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $P \in \mathcal{P}$ such that $|\mathbf{z}| \leq |\mathbf{z}'|$ and $\mathbf{z}(P) > \mathbf{z}'(P)$, there exists $P' \in \mathcal{P}$ such that $\mathbf{z}'(P') > \mathbf{z}(P')$ and $U(\mathbf{z}) + U(\mathbf{z}') \leq U(\mathbf{z} - \mathbf{i}^P + \mathbf{i}^{P'}) + U(\mathbf{z}' + \mathbf{i}^P - \mathbf{i}^{P'})$.*

There is obviously a revenue function version of Theorem OA.1 (again, consider the finest partition).

The combination of the following three local conditions characterize discrete concavity and thus the substitutes condition. The following statement is Theorem 3.2 of Murota (2016).[3]

Theorem OA.2. *A revenue function R is discrete concave if and only if for each $A \subset D$, the following three conditions are all satisfied:*

- *for any distinct $d, d' \in D \setminus A$, $R(A + d + d') + R(A) \leq R(A + d) + R(A + d')$;*
- *for any distinct $d, d', d'' \in D \setminus A$,*

$$R(A + d + d') + R(A + d'') \leq \max\{R(A + d + d'') + R(A + d'), \\ R(A + d) + R(A + d' + d'')\};$$

- *for any distinct $d, d', d'', d''' \in D \setminus A$,*

$$R(A + d + d') + R(A + d'' + d''') \leq \max\{R(A + d + d'') + R(A + d' + d'''), \\ R(A + d + d''') + R(A + d' + d'')\}.$$

[2]The exact statement of the theorem is absent from Murota and Shioura (2018); it combines their Theorems 1.1 and 2.1.

[3]We are unable to track down its first appearance.

OA.2 Proofs

Results are not proved in the order of their appearance in the paper. We often prove more general but later-appearing results before showing that earlier ones are corollaries.

OA.2.1 Proofs of Propositions 10 and 11

In preparation for the main proofs, we establish an equation that a transfer function must satisfy if it preserves the substitutes condition for all spline concave revenue functions in a set of workers (defined in Subsection 4.2).[4]

Lemma OA.3. *If a transfer function T preserves the substitutes condition for all spline concave revenue functions in $A \subset D$, then for any $B \subset D$ not containing $\overline{d}, \underline{d} \in A$, and any $d^* \in D \setminus (A \cup B)$, we have $T(B + \overline{d} + d^*) - T(B + \overline{d}) = T(B + \underline{d} + d^*) - T(B + \underline{d})$.*

Proof. Suppose, for contradiction, that

$$T(B + \overline{d} + d^*) - T(B + \overline{d}) > T(B + \underline{d} + d^*) - T(B + \underline{d}). \tag{OA.1}$$

(The "<" case is symmetric to this ">" case.) In what follows, we will construct a spline concave revenue function R in A such that $R + T$ does not satisfy the substitutes condition.

Let $\alpha \coloneqq \max\{T(C) : C \subset D\} - \min\{T(C) : C \subset D\} + 1 > 0$. We define f such that $f(m) = 3\alpha \min\{m, |A \cap B| + 1\}$ for each $m \in [0, |A|]_{\mathbb{Z}}$, and, accordingly, R such that $R(C) = f(|C \cap A|)$ for each $C \subset D$. Note that because $B \subset D$ does not contain $\overline{d}, \underline{d} \in A$, we can infer that $1 \leq |A \cap B| + 1 \leq |A| - 1$. So R is spline concave in A.

Let $\Delta \coloneqq \frac{1}{2}[(T(B + \overline{d} + d^*) - T(B + \overline{d})) - (T(B + \underline{d} + d^*) - T(B + \underline{d}))] > 0$. Consider a salary schedule $\mathbf{s}$, where $s_{\overline{d}} = \alpha + T(B + \overline{d})$, $s_{\underline{d}} = \alpha + T(B + \underline{d})$, $s_{d^*} = (T(B + \overline{d} + d^*) - T(B + \overline{d})) - \Delta$, $s_d = -4\alpha$ for each $d \in B$, and $s_d = 4\alpha$ for each $d \notin B \cup \{\overline{d}, \underline{d}, d^*\}$.

When the firm faces $\mathbf{s}$, R, and T, we first note that α is so large that all workers in B are demanded, that no worker outside $B \cup \{\overline{d}, \underline{d}, d^*\}$ is demanded, and that either $\overline{d}$ or $\underline{d}$ is demanded (but not both). This leaves four possibilities for demand

[4] To keep the appendix within reasonable length, we have to constantly invoke mathematical objects defined in later sections to prove results in earlier sections.

sets: $B+\overline{d}$, $B+\underline{d}$, $B+\overline{d}+d^*$, and $B+\underline{d}+d^*$. But

$$\begin{aligned}
V(B+\overline{d};\mathbf{s},R+T) &= 3\alpha|B\cap A|+4\alpha|B|+2\alpha;\\
V(B+\underline{d};\mathbf{s},R+T) &= 3\alpha|B\cap A|+4\alpha|B|+2\alpha;\\
V(B+\overline{d}+d^*;\mathbf{s},R+T) &= 3\alpha|B\cap A|+4\alpha|B|+2\alpha+\Delta;\\
V(B+\underline{d}+d^*;\mathbf{s},R+T) &= 3\alpha|B\cap A|+4\alpha|B|+2\alpha-\Delta.
\end{aligned}$$

There is a unique demand set $B+\overline{d}+d^*$.

Now, raise the salary of worker $\overline{d}$ to 4α to obtain a new salary schedule $\mathbf{s}'$. Under $\mathbf{s}'$, the cost of $\overline{d}$ is so large that $B+\overline{d}$ and $B+\overline{d}+d^*$ can never be optimal. We can compare the profits of $B+\underline{d}$ and $B+\underline{d}+d^*$ (which remain the same as before), and conclude that there is a unique demand set $B+\underline{d}$. Therefore, raising the salary of worker $\overline{d}$ excludes worker d^* from the demand set, a violation of the substitutes condition. This contradicts our assumption. □

Under the assumption of Lemma OA.3, for each $B\subset D$ such that $\overline{d},\underline{d}\in A\setminus B$, adding an arbitrary $d^*\in D\setminus(A\cup B)$ to $B+\overline{d}$ brings about the same change in the value of the transfer function T as adding d^* to $B+\underline{d}$. Under the same assumption, we now show that for each $C\subset D$ with $\overline{d}\in A\cap C$ and $\underline{d}\in A\setminus C$, swapping $\overline{d}$ in C for $\underline{d}$ brings about the same change in the value of the transfer function T as swapping $\overline{d}$ in $A\cap C$ for $\underline{d}$.

Lemma OA.4. *If a transfer function T preserves the substitutes condition for all spline concave revenue functions in a fixed $A\subset D$, then for all $C\subset D$, $\overline{d}\in A\cap C$, and $\underline{d}\in A\setminus C$,*

$$T(C-\overline{d}+\underline{d})-T(C)=T((A\cap C)-\overline{d}+\underline{d})-T(A\cap C). \tag{OA.2}$$

Proof. We carry out mathematical induction on $|C\setminus A|$. First, note that the statement is trivially true for each C with $|C\setminus A|=0$; i.e., when $C=A\cap C$. Assume that the statement is true for all C such that $|C\setminus A|\leq m$ with m being a nonnegative integer less than $|D\setminus A|$. Consider $C\subset D$ with $|C\setminus A|=m+1$. There exists $d^*\in C\setminus A$. By the induction hypothesis,

$$T(C-d^*-\overline{d}+\underline{d})-T(C-d^*)=T((A\cap C)-\overline{d}+\underline{d})-T(A\cap C).$$

By Lemma OA.3, we have $T(C-\overline{d}+\underline{d})-T(C-d^*-\overline{d}+\underline{d})=T(C)-T(C-d^*)$ by setting $B=C-d^*-\overline{d}$, or, equivalently,

$$T(C-\overline{d}+\underline{d})-T(C)=T(C-d^*-\overline{d}+\underline{d})-T(C-d^*).$$

We can connect the two equalities above to obtain Equation (OA.2). □

Proof of Proposition 11. Take a transfer function T that preserves the substitutes condition for all within-group binary unit-demand revenue functions. Define an additively separable function T^1 such that $T^1(A) = \sum_{d\in A} T(\{d\})$ for each $A \subset D$. Let $T^2 \coloneqq T - T^1$. We only need to show that T^2 is group concave.

For all $P \in \mathcal{P}$, $d, d' \in P$, and $A \subset D$ such that $d \in A$ and $d' \notin A$, we have

$$\begin{aligned} T^2(A-d+d') - T^2(A) &= (T(A-d+d') - T^1(A-d+d')) - (T(A) - T^1(A)) \\ &= (T(A-d+d') - T(A)) - (T^1(A-d+d') - T^1(A)) \\ &= (T(\{d'\}) - T(\{d\})) - (T(\{d'\}) - T(\{d\})) \\ &= 0, \end{aligned}$$

where the third equality follows from Lemma OA.4[5] and the definition of T^1. To see this, we only need to recognize that the class of all binary unit-demand revenue functions associated with $\overline{d}, \underline{d} \in D$ is the same as the class of all spline concave revenue functions in $\{\overline{d}, \underline{d}\}$, so we can apply Lemma OA.4 by setting $A = \{\overline{d}, \underline{d}\}$.

As a result, for any $A, B \subset D$ with $\tau(A) = \tau(B)$, we can turn A into B by swapping within-group elements one at a time, without changing the transfer specified by T^2, so $T^2(A) = T^2(B)$. In other words, T^2 is group cardinal; i.e., there exists a function $W : \tau(2^D) \to \mathbb{R}$ such that $T^2(A) = W(\tau(A))$ for each $A \subset D$.

The rest of the proof uses a straightforward but lengthy argument to show that T^2 satisfies the substitutes condition. In future proofs, we will refer to it as the "small-α" argument to save space. Suppose T^2 is not group concave. Then, W is not $\mathrm{M}^\natural$-concave by Lemma OA.2. There exist $\overline{\mathbf{z}}, \underline{\mathbf{z}} \in \tau(2^D)$ and $P^* \in \mathcal{P}$ such that $\overline{\mathbf{z}}(P^*) > \underline{\mathbf{z}}(P^*)$, $W(\overline{\mathbf{z}}) + W(\underline{\mathbf{z}}) > W(\overline{\mathbf{z}} - \mathbf{i}^{P^*}) + W(\underline{\mathbf{z}} + \mathbf{i}^{P^*})$, and $W(\overline{\mathbf{z}}) + W(\underline{\mathbf{z}}) > W(\overline{\mathbf{z}} - \mathbf{i}^{P^*} + \mathbf{i}^{P'^*}) + W(\underline{\mathbf{z}} + \mathbf{i}^{P^*} - \mathbf{i}^{P'^*})$ for each $P'^* \in \mathcal{P}$ with $\underline{\mathbf{z}}(P'^*) > \overline{\mathbf{z}}(P'^*)$. Define

$$\begin{aligned} \alpha \coloneqq \frac{1}{4} \min \Big\{ & (W(\overline{\mathbf{z}}) + W(\underline{\mathbf{z}}) - W(\overline{\mathbf{z}} - \mathbf{i}^{P^*}) - W(\underline{\mathbf{z}} + \mathbf{i}^{P^*})), \\ & \min_{P'^* \in \mathcal{P} \text{ with } \underline{\mathbf{z}}(P'^*) > \overline{\mathbf{z}}(P'^*)} (W(\overline{\mathbf{z}}) + W(\underline{\mathbf{z}}) - W(\overline{\mathbf{z}} - \mathbf{i}^{P^*} + \mathbf{i}^{P'^*}) - W(\underline{\mathbf{z}} + \mathbf{i}^{P^*} - \mathbf{i}^{P'^*})) \Big\}, \end{aligned}$$

and consider, for two same-group workers $\overline{d}$ and $\underline{d}$, a binary unit-demand revenue function R with $R(C) = \alpha \min\{1, |C \cap \{\overline{d}, \underline{d}\}|\}$ for each $C \subset D$. We now show that

[5] Specifically, set A and C there to be, respectively, $\{d, d'\}$ and A here, and $\overline{d}$ and $\underline{d}$ there to be d and d' here.

$R+T$ fails the substitutes condition.

Let us consider $A, B \subset D$ such that $\tau(A) = \overline{\mathbf{z}}$, $\tau(B) = \underline{\mathbf{z}}$, and $|A \cap B|$ is the largest possible among those pairs of sets satisfying the last two conditions. Since $\overline{\mathbf{z}}(P^*) > \underline{\mathbf{z}}(P^*)$, there exists $d^* \in P^*$ such that $d^* \in A \setminus B$. We have

$$\begin{aligned}
&(R+T)(A) + (R+T)(B) - (R+T)(A-d^*) - (R+T)(B+d^*) \\
=&(R(A) + R(B) - R(A-d^*) - R(B+d^*)) + \\
&(T^1(A) + T^1(B) - T^1(A-d^*) - T^1(B+d^*)) + \\
&(T^2(A) + T^2(B) - T^2(A-d^*) - T^2(B+d^*)),
\end{aligned}$$

where the first term of the last expression is bounded below by -2α by the construction of R, the second term is 0 by the definition of T^1, and the third term equals $(W(\overline{\mathbf{z}}) + W(\underline{\mathbf{z}}) - W(\overline{\mathbf{z}} - \mathbf{i}^{P^*}) - W(\underline{\mathbf{z}} + \mathbf{i}^{P^*}))$. So the expression is strictly positive, giving us

$$(R+T)(A) + (R+T)(B) > (R+T)(A-d^*) + (R+T)(B+d^*).$$

Further, by the construction of A and B, for each $d'^* \in B \setminus A$, we know the group P'^* containing d'^* satisfies $\underline{\mathbf{z}}(P'^*) > \overline{\mathbf{z}}(P'^*)$; so we can similarly show

$$(R+T)(A) + (R+T)(B) > (R+T)(A-d^*+d'^*) + (R+T)(B+d^*-d'^*).$$

In conclusion, $R+T$ cannot be discrete concave. By Lemma OA.1, $R+T$ fails the substitutes condition, a contradiction. □

Proposition 10 is a corollary of Proposition 11.

Proof of Proposition 10. Take a transfer function T that preserves the substitutes condition for all binary unit-demand revenue functions. Let the partition be the coarsest: $\mathcal{P} = \{D\}$. Then the set of all within-group binary unit-demand revenue functions is the set of all binary unit-demand revenue functions. Applying Proposition 11 tells us that $T = T^1 + T^2$, where T^1 is additively separable, and T^2 is group concave. But given $\mathcal{P} = \{D\}$, group concavity is the same as cardinal concavity. □

OA.2.2 Proof of Theorem 1

We first make a simple observation that the sum of two transfer functions that always preserve the substitutes condition still always preserves the substitutes condition.[6]

[6]Note that the class of transfer functions that preserve the substitutes condition for a subclass of revenue functions may not be closed under addition.

Lemma OA.5. *The class of transfer functions that always preserve the substitutes condition is closed under addition.*

Proof. Consider any two transfer functions T^1 and T^2 that always preserve the substitutes condition. We need to show that $T^1 + T^2$ preserves the substitutes condition. But given any revenue function R satisfying the substitutes condition, by the definition of always preserving the substitutes condition, $R+T^1$ satisfies the condition, and then $R+(T^1+T^2) = (R+T^1)+T^2$ also satisfies the condition. □

The proof of Theorem 1 utilizes results from KSY, so we need notations about constraints listed in Subsection 6.2. A classical characterization of the substitutes condition by Gul and Stacchetti (1999) generalizes to the case with constraints. A demand correspondence $X(\,\cdot\,; R, \mathcal{F})$ satisfies the **single-improvement property** if for all salary schedule $\mathbf{s}$ and $A \in \mathcal{F}$ such that $A \notin X(\mathbf{s}; R, \mathcal{F})$, there exists $A' \in \mathcal{F}$ such that $V(A; \mathbf{s}, R) < V(A'; \mathbf{s}, R)$, $|A \setminus A'| \leq 1$, and $|A' \setminus A| \leq 1$.

Lemma OA.6 (Gul and Stacchetti (1999))**.** *A demand correspondence $X(\,\cdot\,; R, \mathcal{F})$ satisfies the substitutes condition if and only if it satisfies the single-improvement property.*

A feasibility collection $\mathcal{F}$ **always preserves the substitutes condition** if for each revenue function R satisfying the condition, $X(\,\cdot\,; R, \mathcal{F})$ satisfies it. Theorem 1 of KSY implies that a feasibility collection defined by an interval constraint always preserves the substitutes condition. Now we proceed to prove Theorem 1.

Proof of Theorem 1. The necessity part follows from Proposition 10. For sufficiency, due to Proposition 1 and Lemma OA.5, we only need to show that a cardinally concave transfer function T always preserves the substitutes condition, i.e., given a revenue function R satisfying the substitutes condition, $R + T$ satisfies it too. By Lemma OA.6, we can instead prove the single-improvement property of $R+T$ (when the firm is unconstrained with $\mathcal{F} = 2^D$).

Consider a salary schedule $\mathbf{s}$ and a set of workers A such that $A \notin X(\mathbf{s}; R+T, 2^D)$; i.e., there exists $A^* \subset D$ such that

$$R(A^*) + T(A^*) - \sum_{d \in A^*} s_d > R(A) + T(A) - \sum_{d \in A} s_d. \tag{OA.3}$$

We need to find $A' \subset D$ such that $|A \setminus A'| \leq 1$, $|A' \setminus A| \leq 1$, and $R(A') + T(A') - \sum_{d \in A'} s_d > R(A) + T(A) - \sum_{d \in A} s_d$.

Let f be the concave-extensible function on $[0, M]_{\mathbb{Z}}$ associated with T. Define $\overline{\alpha} \coloneqq f(|A|) - f(|A| - 1)$ and $\underline{\alpha} \coloneqq f(|A| + 1) - f(|A|)$. Construct transfer functions $\overline{T}$ and $\underline{T}$ such that $\overline{T}(B) = f(|A|) + \overline{\alpha}(|B| - |A|)$ and $\underline{T}(B) = f(|A|) + \underline{\alpha}(|B| - |A|)$ for each $B \subset D$. By the concave extensibility of f, we have $\overline{T} \geq T$ and $\underline{T} \geq T$.

Note that $\overline{T}$ and $\underline{T}$ are cardinal, each equal to a constant plus an additively separable function (which assigns the same transfer amount to each worker). Since a constant transfer function does not change demand correspondences, it always preserves the substitutes condition. Hence, by Proposition 1 and Lemma OA.5, $\overline{T}$ and $\underline{T}$ both always preserve the substitutes condition. In particular, $R + \overline{T}$ and $R + \underline{T}$ satisfy the substitutes condition.

When $|A^*| \leq |A|$, we have

$$\begin{aligned} R(A^*) + \overline{T}(A^*) - \sum_{d \in A^*} s_d \; &\geq R(A^*) + T(A^*) - \sum_{d \in A^*} s_d \\ &> R(A) + T(A) - \sum_{d \in A} s_d = \; R(A) + \overline{T}(A) - \sum_{d \in A} s_d, \end{aligned}$$

where the first inequality follows from $\overline{T} \geq T$, and the second from Inequality (OA.3). So A is less profitable than A^* under $\mathbf{s}$ and $R + \overline{T}$. Theorem 1 of KSY implies that, as $R + \overline{T}$ satisfies the substitutes condition, $X(\,\cdot\,; R + \overline{T}, \mathcal{D}_{[0,|A|]})$ does too. But note that A^* is feasible under the ceiling constraint of $|A|$ by assumption. Lemma OA.6 and the suboptimality of A imply the existence of $A' \in \mathcal{D}_{[0,|A|]}$ such that $|A' \backslash A| \leq 1$, $|A \backslash A'| \leq 1$, and $R(A') + \overline{T}(A') - \sum_{d \in A'} s_d > R(A) + \overline{T}(A) - \sum_{d \in A} s_d$. But then

$$\begin{aligned} R(A') + T(A') - \sum_{d \in A'} s_d \; &= R(A') + \overline{T}(A') - \sum_{d \in A'} s_d \\ &> R(A) + \overline{T}(A) - \sum_{d \in A} s_d = \; R(A) + T(A) - \sum_{d \in A} s_d, \end{aligned}$$

where the first equality follows from the fact that $|A'|$ equals $|A|$ or $|A| - 1$.

The case of $|A^*| > |A|$ is analogous. Theorem 1 of KSY implies that as $R + \underline{T}$ satisfies the substitutes condition, $X(\,\cdot\,; R + \underline{T}, \mathcal{D}_{[|A|,M]})$ does too. A single-improvement opportunity A' over A under the floor constraint is a single-improvement opportunity without the constraint. We skip the details. □

OA.2.3 Proof of Proposition 13

We first introduce a simple consequence of Lemma OA.3.

Lemma OA.7. *If a vectorial function W preserves the vectorial substitutes*

condition for all spline concave vectorial functions in a fixed $Q = \cup\mathcal{Q}$, where $\mathcal{Q} \subsetneq \mathcal{P}$, then for all $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $P^* \in \mathcal{P} \setminus \mathcal{Q}$ such that $|\mathbf{z}(\mathcal{Q})| = |\mathbf{z}'(\mathcal{Q})|$, $\mathbf{z}(P) = \mathbf{z}'(P)$ for each $P \in \mathcal{P} \setminus \mathcal{Q}$, and $\mathbf{z}(P^*) < |P^*|$, we have

$$W(\mathbf{z} + \mathbf{i}^{P^*}) - W(\mathbf{z}) = W(\mathbf{z}' + \mathbf{i}^{P^*}) - W(\mathbf{z}'). \tag{OA.4}$$

Proof. Let W be associated with a group concave transfer function T. Then T preserves the substitutes condition for all spline concave revenue functions in Q. Because $|\mathbf{z}(\mathcal{Q})| = |\mathbf{z}'(\mathcal{Q})|$ and $\mathbf{z}(P) = \mathbf{z}'(P)$ for each $P \in \mathcal{P} \setminus \mathcal{Q}$, we can find $A, A' \subset D$ such that $\tau(A) = \mathbf{z}$, $\tau(A') = \mathbf{z}'$, $|A \cap Q| = |A' \cap Q|$, and $A \setminus Q = A' \setminus Q$. Let $d^* \in P^* \setminus A$.

Lemma OA.3 tells us that in the case of $|A \setminus A'| = |A' \setminus A| = 1$, $T(A+d^*) - T(A) = T(A'+d^*) - T(A')$, which implies Equation (OA.4). In the case of $|A \setminus A'| = |A' \setminus A| > 1$, we can turn A into A' one element at a time, obtain a series of equations, and connect them to obtain $T(A+d^*) - T(A) = T(A'+d^*) - T(A')$. So Equation (OA.4) holds. □

Equation (OA.4) helps us dissect W.

Lemma OA.8. *When $|\mathcal{P}| > 2$, for a vectorial function W, if Equation (OA.4) holds for all $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $P^* \in \mathcal{P}$ such that $|\mathbf{z}(\mathcal{P} - P^*)| = |\mathbf{z}'(\mathcal{P} - P^*)|$, $\mathbf{z}(P^*) = \mathbf{z}'(P^*) < |P^*|$, then there exists a family of real-valued functions on $[0, M]_{\mathbb{Z}}$, $\{f_P\}_{P \in \mathcal{P}}$, and $f : [0, M]_{\mathbb{Z}} \to \mathbb{R}$ such that $W(\mathbf{z}) = \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$.*

Proof. We assign values to $\{f_P\}_{P \in \mathcal{P}}$ and f by induction. Let $f(0) = W(\tau(\emptyset))$, $f(1) = 0$, and for each $P \in \mathcal{P}$, $f_P(0) = 0$ and $f_P(1) = W(\mathbf{i}^P)$. Further, for $p \in [2, |\mathcal{P}|]_{\mathbb{Z}}$, set $f(p) = W(\mathbf{z}) - \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P))$ for all $\mathbf{z} \in \tau(2^D)$ with $|\mathbf{z}| = p$ and $\mathbf{z} \leq \sum_{P \in \mathcal{P}} \mathbf{i}^P$. We need to show that this way of assigning values to f is independent of the chosen $\mathbf{z}$. We carry out induction on $|\mathbf{z}|$.

The base cases of $|\mathbf{z}| = 1$ is trivial. Assume that for some $p \in [1, |\mathcal{P}| - 1]_{\mathbb{Z}}$, $f(p) = W(\mathbf{z}) - \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P))$ is independent of the chosen $\mathbf{z} \in \tau(2^D)$ with $|\mathbf{z}| = p$ and $\mathbf{z} \leq \sum_{P \in \mathcal{P}} \mathbf{i}^P$. Consider $\overline{\mathbf{z}}, \underline{\mathbf{z}} \in \tau(2^D)$ with $|\overline{\mathbf{z}}| = |\underline{\mathbf{z}}| = p + 1$ and $\overline{\mathbf{z}}, \underline{\mathbf{z}} \leq \sum_{P \in \mathcal{P}} \mathbf{i}^P$. First, impose the additional assumption that there exists $P^* \in \mathcal{P}$

with $\overline{\mathbf{z}}(P^*) = \underline{\mathbf{z}}(P^*) = 1$. Hence, we have

$$\begin{aligned}
&\Big(W(\overline{\mathbf{z}}) - \sum_{P\in\mathcal{P}} f_P(\overline{\mathbf{z}}(P))\Big) - \Big(W(\underline{\mathbf{z}}) - \sum_{P\in\mathcal{P}} f_P(\underline{\mathbf{z}}(P))\Big) \\
=&\,(W(\overline{\mathbf{z}}) - W(\underline{\mathbf{z}})) - \sum_{P\in\mathcal{P}} (f_P(\overline{\mathbf{z}}(P)) - f_P(\underline{\mathbf{z}}(P))) \\
=&\,(W(\overline{\mathbf{z}} - \mathbf{i}^{P^*}) - W(\underline{\mathbf{z}} - \mathbf{i}^{P^*})) - \sum_{P\in\mathcal{P}} (f_P(\overline{\mathbf{z}}(P)) - f_P(\underline{\mathbf{z}}(P))) = 0,
\end{aligned}$$

where the second equality follows from Equation (OA.4) and the third follows from the inductive hypothesis. So the assignment of $f(p+1)$ is the same based on any two $\overline{\mathbf{z}}$ and $\underline{\mathbf{z}}$ that share a positive dimension. But any $\overline{\mathbf{z}}, \underline{\mathbf{z}} \in \tau(2^D)$ with $|\overline{\mathbf{z}}| = |\underline{\mathbf{z}}| > 1$ and $\overline{\mathbf{z}}, \underline{\mathbf{z}} \leq \sum_{P\in\mathcal{P}} \mathbf{i}^P$ are connected through a series of such $\mathbf{z} \in \tau(2^D)$, where any two neighbors share a positive dimension. We can conclude that the assignment of $f(p+1)$ is the same based on any two $\overline{\mathbf{z}}$ and $\underline{\mathbf{z}}$, and finish the induction.

Given the above, we can start another induction with the base case as follows: $W(\mathbf{z}) = \sum_{P\in\mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$ with $\mathbf{z} \leq \sum_{P\in\mathcal{P}} \mathbf{i}^P$. Assume that we have assigned values such that $W(\mathbf{z}) = \sum_{P\in\mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$ with $\mathbf{z} \leq \mathbf{y}$, where $\mathbf{y} \in \tau(2^D)$, $\mathbf{y} \geq \sum_{P\in\mathcal{P}} \mathbf{i}^P$, and $\mathbf{y}(\overline{P}) < |\overline{P}|$ for some $\overline{P} \in \mathcal{P}$. Assign new values such that

$$\begin{aligned}
f_{\overline{P}}(\mathbf{y}(\overline{P}) + 1) &= W((\mathbf{y}(\overline{P}) + 1)\mathbf{i}^{\overline{P}}) - f(\mathbf{y}(\overline{P}) + 1); \\
f(|\mathbf{y}| + 1) &= W(\mathbf{y} + \mathbf{i}^{\overline{P}}) - \sum_{P\in\mathcal{P}} f_P((\mathbf{y} + \mathbf{i}^{\overline{P}})(P)).
\end{aligned}$$

We only need to show that $W(\mathbf{z}) = \sum_{P\in\mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$ with $\tau(\emptyset) \leq \mathbf{z} \leq \mathbf{y} + \mathbf{i}^{\overline{P}}$ and $\mathbf{z}(\overline{P}) = \mathbf{y}(\overline{P}) + 1$. We do this by strong induction on $m := |\mathbf{z}(\mathcal{P} - \overline{P})|$.[7] The case of $m = 0$ follows from the assignment of $f_{\overline{P}}(\mathbf{y}(\overline{P})+1)$ and $f_P(0)$ above. The case of $m = |\mathbf{y}(\mathcal{P} - \overline{P})|$ follows from the assignment of $f(|\mathbf{y}| + 1)$.

Assume that for some $m \in [0, |\mathbf{y}(\mathcal{P} - \overline{P})| - 2]_{\mathbb{Z}}$, $W(\mathbf{z}) = \sum_{P\in\mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$ satisfying $\mathbf{z}(\overline{P}) = \mathbf{y}(\overline{P})+1$, $\mathbf{z}(P) \leq \mathbf{y}(P)$ for each $P \in \mathcal{P} - \overline{P}$, and $|\mathbf{z}(\mathcal{P} - \overline{P})| \leq m$. Consider $\mathbf{x} \in \tau(2^D)$ that satisfies $\mathbf{x}(\overline{P}) = \mathbf{y}(\overline{P}) + 1$, $\mathbf{x}(P) \leq \mathbf{y}(P)$ for each $P \in \mathcal{P} - \overline{P}$, and $|\mathbf{x}(\mathcal{P} - \overline{P})| = m + 1$. Since $m + 1 \in [1, |\mathbf{y}(\mathcal{P} - \overline{P})| - 1]_{\mathbb{Z}}$, $|\mathcal{P}| > 2$, and $\mathbf{y} \geq \sum_{P\in\mathcal{P}} \mathbf{i}^P$, there exists distinct $P^*, \underline{P} \in \mathcal{P} - \overline{P}$ such that $\mathbf{x}(P^*) \geq 1$

[7] Note that this is an induction argument within an induction argument.

and $\mathbf{x}(\underline{P}) < \mathbf{y}(\underline{P})$. Hence, we have

$$\begin{aligned} W(\mathbf{x}) &= W(\mathbf{x} - \mathbf{i}^{P^*}) + W(\mathbf{x} - \mathbf{i}^{\overline{P}} + \mathbf{i}^{\underline{P}}) - W(\mathbf{x} - \mathbf{i}^{\overline{P}} - \mathbf{i}^{P^*} + \mathbf{i}^{\underline{P}}) \\ &= \sum_{P \in \mathcal{P}} f_P(\mathbf{x}(P)) + f(|\mathbf{x}|), \end{aligned}$$

where the first equality follows from Equation (OA.4), and the second follows from the induction hypotheses, i.e., $W(\mathbf{z}) = \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ holds for all three terms.

In conclusion, we have $W(\mathbf{z}) = \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$ satisfying $\mathbf{z}(\overline{P}) = \mathbf{y}(\overline{P}) + 1$, $\mathbf{z}(P) \leq \mathbf{y}(P)$ for $P \in \mathcal{P} - \overline{P}$, and $|\mathbf{z}(\mathcal{P} - \overline{P})| \leq |\mathbf{y}(\mathcal{P} - \overline{P})| - 1$. Combining this with earlier results, we are done. □

Proof of Proposition 13. Take a vectorial function W on $\tau(2^D)$ that preserves the vectorial substitutes condition for all V-$\overline{\text{uni}}$-group spline concave vectorial functions. By Lemma OA.7, we know the conditions of Lemma OA.8 are satisfied. We only need to show that all obtained functions, $\{f_P\}_{P \in \mathcal{P}}$ and f such that $W(\mathbf{z}) = \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$, are concave-extensible.

Let $\alpha_P(m+1) = f_P(m+1) - f_P(m)$ for all $m \in [0, |P|-1]_{\mathbb{Z}}$ and $P \in \mathcal{P}$. Lemma OA.8 says that for all distinct $P, P' \in \mathcal{P}$ and $m \in [1, |P|-1]_{\mathbb{Z}}$,

$$W((m+1)\mathbf{i}^P) - W(m\mathbf{i}^P + \mathbf{i}^{P'}) = f_P(m+1) - f_P(m) - f_{P'}(1) = \alpha_P(m+1) - f_{P'}(1);$$
$$W(m\mathbf{i}^P) - W((m-1)\mathbf{i}^P + \mathbf{i}^{P'}) = f_P(m) - f_P(m-1) - f_{P'}(1) = \alpha_P(m) - f_{P'}(1).$$

Using a small-α argument analogous to the one in the proof of Proposition 11, we can show that $W : \tau(2^D) \to \mathbb{R}$ must be M$^\natural$-concave. Thus, by the first condition in Theorem OA.1, we have

$$\begin{aligned} &\alpha_P(m+1) - \alpha_P(m) \\ =& (W((m+1)\mathbf{i}^P) + W((m-1)\mathbf{i}^P + \mathbf{i}^{P'})) - (W(m\mathbf{i}^P + \mathbf{i}^{P'}) + W(m\mathbf{i}^P)) \\ \leq& (W(m\mathbf{i}^P) + W(m\mathbf{i}^P + \mathbf{i}^{P'})) - (W(m\mathbf{i}^P + \mathbf{i}^{P'}) + W(m\mathbf{i}^P)) \;=\; 0. \end{aligned}$$

Therefore, f_P is concave-extensible.

Let $\beta(m+1) = f(m+1) - f(m)$ for $m \in [0, |D|-1]_{\mathbb{Z}}$. For each $m \in [1, |D|-1]_{\mathbb{Z}}$, find $\mathbf{z} \in \tau(2^D)$ and $\overline{P}, \underline{P} \in \mathcal{P}$ such that $\mathbf{z}(\overline{P}), \mathbf{z}(\underline{P}) \geq 1$ and $|\mathbf{z}| = m + 1 \geq 2$. By

Lemma OA.8, we know

$$\begin{aligned} W(\mathbf{z}) - W(\mathbf{z} - \mathbf{i}^{\overline{P}}) &= [f(m+1) - f(m)] + [f_{\overline{P}}(\mathbf{z}(\overline{P})) - f_{\overline{P}}(\mathbf{z}(\overline{P}) - 1)] \\ &= \beta(m+1) + [f_{\overline{P}}(\mathbf{z}(\overline{P})) - f_{\overline{P}}(\mathbf{z}(\overline{P}) - 1)]; \\ W(\mathbf{z} - \mathbf{i}^{\underline{P}}) - W(\mathbf{z} - \mathbf{i}^{\overline{P}} - \mathbf{i}^{\underline{P}}) &= [f(m) - f(m-1)] + [f_{\overline{P}}(\mathbf{z}(\overline{P})) - f_{\overline{P}}(\mathbf{z}(\overline{P}) - 1)] \\ &= \beta(m) + [f_{\overline{P}}(\mathbf{z}(\overline{P})) - f_{\overline{P}}(\mathbf{z}(\overline{P}) - 1)]. \end{aligned}$$

Thus, by the first condition in Theorem OA.1, we have

$$\begin{aligned} \beta(m+1) - \beta(m) &= [W(\mathbf{z}) + W(\mathbf{z} - \mathbf{i}^{\overline{P}} - \mathbf{i}^{\underline{P}})] - [W(\mathbf{z} - \mathbf{i}^{\overline{P}}) + W(\mathbf{z} - \mathbf{i}^{\underline{P}})] \\ &\leq [W(\mathbf{z} - \mathbf{i}^{\overline{P}}) + W(\mathbf{z} - \mathbf{i}^{\underline{P}})] - [W(\mathbf{z} - \mathbf{i}^{\overline{P}}) + W(\mathbf{z} - \mathbf{i}^{\underline{P}})] = 0. \end{aligned}$$

Therefore, f is concave-extensible. □

OA.2.4 Proof of Proposition 12

Lemma OA.4 easily implies the following results.

Lemma OA.9. *If a transfer function T preserves the substitutes condition for all spline concave revenue functions in a fixed $A \subset D$, then for all $B \subset D \setminus A$ and $\overline{C}, \underline{C} \subset A$ such that $|\overline{C}| = |\underline{C}|$,*

$$T(B \cup \overline{C}) - T(B \cup \underline{C}) = T(\overline{C}) - T(\underline{C}). \tag{OA.5}$$

Proof. We can start with setting the C in Equation (OA.2) to be $B \cup \underline{C}$, and swapping one element in $\underline{C} \setminus \overline{C}$–say $\underline{d}$–for one element in $\overline{C} \setminus \underline{C}$–say $\overline{d}$. This gives us one equation. Since $|\overline{C}| = |\underline{C}|$, we can continue the swapping until $\underline{C}$ turns into $\overline{C}$, and connect all obtained equations to reach Equation (OA.5). □

Proof of Proposition 12. Take a transfer function T that preserves the substitutes condition for all uni-group and $\overline{\text{uni}}$-group spline concave revenue functions. Define a transfer function T^1 such that $T^1(A) = \sum_{P \in \mathcal{P}} T(A \cap P)$ for each $A \subset D$, and let $T^2 \coloneqq T - T^1$.

For all $P \in \mathcal{P}$, $d, d' \in P$, and $A \subset D$ such that $d \in A$ and $d' \notin A$, we have

$$\begin{aligned} &T^2(A - d + d') - T^2(A) \\ =& (T(A - d + d') - T^1(A - d + d')) - (T(A) - T^1(A)) \\ =& (T(A - d + d') - T(A)) - (T^1(A - d + d') - T^1(A)) \\ =& (T(A - d + d') - T(A)) - (T((A \cap P) - d + d') - T(A \cap P)) = 0, \end{aligned}$$

where the third equality follows from the definition of T^1, and the fourth follows from Lemma OA.9. As a result, T^2 is group cardinal; there exists a vectorial function W on $\tau(2^D)$ such that for each $A \subset D$, $T^2(A) = W(\tau(A))$. We have $T(A) = \sum_{P\in\mathcal{P}} T(A \cap P) + W(\tau(A))$ for each $A \subset D$.

We now show that W satisfies the conditions of Lemma OA.8. Pick any $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $P^* \in \mathcal{P}$ such that $|\mathbf{z}(\mathcal{P} - P^*)| = |\mathbf{z}'(\mathcal{P} - P^*)|$, $\mathbf{z}(P^*) = \mathbf{z}'(P^*) < |P^*|$. We can find $A, A' \subset D$ such that $\tau(A) = \mathbf{z}$, $\tau(A') = \mathbf{z}'$, and $A \cap P^* = A' \cap P^*$. Because T preserves the substitutes condition for all $\overline{\text{uni}}$-group spline concave revenue functions, by Lemma OA.9, for each $d \in P^* \setminus A$, we can derive

$$T(A + d) - T(A) = T(A' + d) - T(A'). \tag{OA.6}$$

We obtain Equation (OA.4) through

$$\begin{aligned}
W(\mathbf{z} + \mathbf{i}^{P^*}) - W(\mathbf{z}) &= W(\tau(A + d)) - W(\tau(A)) \\
&= (T(A + d) - T^1(A + d)) - (T(A) - T^1(A)) \\
&= (T(A + d) - T(A)) - (T((A + d) \cap P^*) - T(A \cap P^*)) \\
&= (T(A' + d) - T(A')) - (T((A' + d) \cap P^*) - T(A' \cap P^*)) \\
&= (T(A' + d) - T^1(A' + d)) - (T(A') - T^1(A')) \\
&= W(\tau(A' + d)) - W(\tau(A')) \\
&= W(\mathbf{z}' + \mathbf{i}^{P^*}) - W(\mathbf{z}'),
\end{aligned}$$

where the fourth equality follows from $A \cap P^* = A' \cap P^*$ and Equation (OA.6). Therefore, by Lemma OA.8, there exists $\{f_P\}_{P\in\mathcal{P}}$ and f such that $W(\mathbf{z}) = \sum_{P\in\mathcal{P}} f_P(\mathbf{z}(P)) + f(|\mathbf{z}|)$ for each $\mathbf{z} \in \tau(2^D)$. For each $P \in \mathcal{P}$, define $T_P : 2^P \to \mathbb{R}$ such that $T_P(A) = T(A) + f_P(|A|)$ for each $A \subset P$. We have $T(A) = \sum_{P\in\mathcal{P}} T_P(A \cap P) + f(|A|)$ for each $A \subset D$.

Next, we show that each T_P satisfies the substitutes condition on 2^P, through proving the three conditions of Theorem OA.2. Let $\Psi \coloneqq \sum_{P'\in\mathcal{P}-P} T_{P'}(\emptyset)$. Pick any $A, B \subset P$ with $|A| = |B|$ and any $d \in A \setminus B$. Using a small-α argument analogous to the one in the proof of Proposition 11, we can prove that T must satisfy the substitutes condition. By the second condition in the revenue function version of Theorem OA.1, there exists $d' \in B \setminus A$ such that

$$T(A) + T(B) \leq T(A - d + d') + T(B + d - d').$$

Then we know the third condition of Theorem OA.2 holds as a special case of the

following (i.e., when $|A \setminus B| = 2$):

$$\begin{aligned}
&T_P(A) + T_P(B) \\
=&(T(A) - f(|A|) - \Psi) + (T(B) - f(|B|) - \Psi) \\
\leq&(T(A - d + d') - f(|A - d + d'|) - \Psi) + (T(B + d - d') - f(|B + d - d'|) - \Psi) \\
=&T_P(A - d + d') + T_P(B + d - d').
\end{aligned}$$

Pick any $A \subset P$ and distinct $d, d', d'' \in P \setminus A$. By the first condition in the revenue function version of Theorem OA.1, we have

$$T(A+d+d')+T(A+d'') \leq \max\{T(A+d)+T(A+d'+d''),\, T(A+d')+T(A+d+d'')\}.$$

Without loss of generality, assume $T(A+d+d')+T(A+d'') \leq T(A+d)+T(A+d'+d'')$. Then we know the second condition of Theorem OA.2 holds:

$$\begin{aligned}
&T_P(A + d + d') + T_P(A + d'') \\
=&(T(A + d + d') - f(|A| + 2) - \Psi) + (T(A + d'') - f(|A| + 1) - \Psi) \\
\leq&(T(A + d' + d'') - f(|A| + 2) - \Psi) + (T(A + d) - f(|A| + 1) - \Psi) \\
=&T_P(A + d' + d'') + T_P(A + d).
\end{aligned}$$

Pick any $A \subset P$ and any distinct $d, d' \in P \setminus A$. As $|\mathcal{P}| \geq 3$, choose any two distinct $\bar{P}, \hat{P} \in \mathcal{P} - P$, any $d'' \in \bar{P}$, and $d''' \in \hat{P}$. By the second condition in the revenue function version of Theorem OA.1, we see that

$$\begin{aligned}
&T(A + d + d') + T(A + d'' + d''') \\
&\quad \leq \max\{T(A + d + d'') + T(A + d' + d'''),\, T(A + d' + d'') + T(A + d + d''')\}.
\end{aligned}$$

Without loss of generality, assume $T(A + d + d') + T(A + d'' + d''') \leq T(A + d + d'') + T(A + d' + d''')$. Then we know the first condition of Theorem OA.2 holds:

$$\begin{aligned}
&T_P(A + d + d') + T_P(A) \\
=\ &(T(A + d + d') - f(|A| + 2) - \Psi) \\
&\quad + (T(A + d'' + d''') - f(|A| + 2) - \Psi + T_{\bar{P}}(\emptyset) + T_{\hat{P}}(\emptyset) - T_{\bar{P}}(d'') - T_{\hat{P}}(d''')) \\
\leq\ &(T(A + d + d'') - f(|A| + 2) - \Psi + T_{\bar{P}}(\emptyset) - T_{\bar{P}}(d'')) \\
&\quad + (T(A + d' + d''') - f(|A| + 2) - \Psi + T_{\hat{P}}(\emptyset) - T_{\hat{P}}(d''')) \\
=\ &T_P(A + d) + T_P(A + d').
\end{aligned}$$

Finally, we prove f is concave-extensible. For each $m \in [1, |D| - 1]_{\mathbb{Z}}$, pick $A \subset D$ such that $|A| = m - 1$ and there exist $d', d'' \in D \setminus A$ which belong

to distinct $P', P'' \in \mathcal{P}$ respectively. Let $\Phi \coloneqq \sum_{P\in\mathcal{P}} T_P(A \cap P)$. Note that $T(A + d' + d'') + T(A) \leq T(A + d') + T(A + d'')$ by the first condition of Theorem OA.2. Thus we know

$$
\begin{aligned}
&f(m+1) - f(m) \\
&= (T(A + d' + d'') - \Phi + T_{P'}(A \cap P') + T_{P''}(A \cap P'') - T_{P'}((A \cap P') + d') \\
&\qquad - T_{P''}((A \cap P'') + d'')) \ - \ (T(A + d') - \Phi + T_{P'}(A \cap P') - T_{P'}((A \cap P') + d')) \\
&\leq (T(A + d'') - \Phi + T_{P''}(A \cap P'') - T_{P''}((A \cap P'') + d'')) \ - \ (T(A) - \Phi) \\
&= f(m) - f(m-1).
\end{aligned}
$$

This concludes the proof. □

OA.2.5 Proofs of Theorems 2, 3, and 6

We prove a known result for completeness (Murota, 2021b, the third statement of Proposition 4.14).

Lemma OA.10. *A V-additively separable function always preserves the vectorial substitutes condition.*

Proof. Let a vectorial function U satisfy the vectorial substitutes condition and thus $\mathrm{M}^\natural$-concavity (Lemma OA.2), and a vectorial function W be V-additively separable with respect to a family of concave-extensible functions $\{f_P\}_{P\in\mathcal{P}}$. Take any $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $\hat{P} \in \mathcal{P}$ such that $\mathbf{z}(\hat{P}) > \mathbf{z}'(\hat{P})$. Since U is $\mathrm{M}^\natural$-concave, either $U(\mathbf{z}-\mathbf{i}^{\hat{P}})+U(\mathbf{z}'+\mathbf{i}^{\hat{P}}) \geq U(\mathbf{z})+U(\mathbf{z}')$, or there exists $\hat{P}' \in \mathcal{P}$ such that $\mathbf{z}'(\hat{P}') > \mathbf{z}(\hat{P}')$ and $U(\mathbf{z} - \mathbf{i}^{\hat{P}} + \mathbf{i}^{\hat{P}'}) + U(\mathbf{z}' + \mathbf{i}^{\hat{P}} - \mathbf{i}^{\hat{P}'}) \geq U(\mathbf{z}) + U(\mathbf{z}')$, or both.

In the first case, $U(\mathbf{z}-\mathbf{i}^{\hat{P}})+U(\mathbf{z}'+\mathbf{i}^{\hat{P}})-U(\mathbf{z})-U(\mathbf{z}')$ is nonnegative. Moreover, by the V-additive separability of W and $\mathbf{z}(\hat{P}) > \mathbf{z}'(\hat{P})$,

$$
\begin{aligned}
&W(\mathbf{z} - \mathbf{i}^{\hat{P}}) + W(\mathbf{z}' + \mathbf{i}^{\hat{P}}) - W(\mathbf{z}) - W(\mathbf{z}') \\
&= [W(\mathbf{z} - \mathbf{i}^{\hat{P}}) - W(\mathbf{z})] + [W(\mathbf{z}' + \mathbf{i}^{\hat{P}}) - W(\mathbf{z}')] \\
&= [f_{\hat{P}}(\mathbf{z}(\hat{P}) - 1) - f_{\hat{P}}(\mathbf{z}(\hat{P}))] + [f_{\hat{P}}(\mathbf{z}'(\hat{P}) + 1) - f_{\hat{P}}(\mathbf{z}'(\hat{P}))] \geq 0.
\end{aligned}
$$

So we have $(U + W)(\mathbf{z} - \mathbf{i}^{\hat{P}}) + (U + W)(\mathbf{z}' + \mathbf{i}^{\hat{P}}) - (U + W)(\mathbf{z}) - (U + W)(\mathbf{z}') \geq 0$.

In the second case, we can similarly show

$$
(U + W)(\mathbf{z} - \mathbf{i}^{\hat{P}} + \mathbf{i}^{\hat{P}'}) + (U + W)(\mathbf{z}' + \mathbf{i}^{\hat{P}} - \mathbf{i}^{\hat{P}'}) \geq (U + W)(\mathbf{z}) + (U + W)(\mathbf{z}').
$$

Combining these two cases proves that $U + W$ is $\mathrm{M}^\natural$-concave. □

Analogous to Lemma OA.5, it is easy to see the following result.

Lemma OA.11. *The class of vectorial functions that always preserve the vectorial substitutes condition is closed under addition.*

Proof of Theorem 6. The necessity part follows from Proposition 13. For sufficiency, in view of lemmata OA.10 and OA.11, we only need to show that given a vectorial function U associated with a revenue function R satisfying the substitutes condition, and a vectorial function W associated with a cardinally concave transfer function T, $U + W$ satisfies the vectorial substitutes condition. But Theorem 1 says that $R + T$ satisfies the substitutes condition. Because $U + W$ is associated with $R + T$, it satisfies the vectorial substitutes condition. □

Adding group separable transfer functions to group concave revenue functions gives us a new subclass of functions satisfying the substitutes condition.

Lemma OA.12. *The sum of a group concave revenue function and a group separable transfer function satisfies the substitutes condition.*

Proof. We show that $R + T$, where R is a group separable revenue function and T is a group concave transfer function, satisfies the single-improvement property, that is, for all salary schedule $\mathbf{s}$ and $A \subset D$ such that $A \notin X(\mathbf{s}; R + T)$, we need to find $A' \subset D$ such that $V(A; \mathbf{s}, R + T) < V(A'; \mathbf{s}, R + T)$, $|A \setminus A'| \leq 1$, and $|A' \setminus A| \leq 1$.

Within each group P, denote the profit function associated with its revenue function R_P by $V_P(\,\cdot\,; \mathbf{s}|_P, R_P)$.[8] Suppose that there exists $P^* \in \mathcal{P}$ and $d, d' \in P^*$ such that $d \in A$, $d' \notin A$, and $V_{P^*}((A \cap P^*) - d + d'; \mathbf{s}|_{P^*}, R_{P^*}) > V_{P^*}(A \cap P^*; \mathbf{s}|_{P^*}, R_{P^*})$. In such a case, because $V(A - d + d'; \mathbf{s}, R) > V(A; \mathbf{s}, R)$ and $T(A - d + d') = T(A)$, we have $V(A - d + d'; \mathbf{s}, R + T) > V(A; \mathbf{s}, R + T)$. We can set $A' = A - d + d'$ for this simple case. Thus, without loss of generality, we assume nonexistence of such P^*.

Hence, within every group $P \in \mathcal{P}$, $A \cap P$ offers the maximal profit under the exact constraint of $|A \cap P|$ (given the salary schedule $\mathbf{s}|_P$ and revenue function R_P). For every $P \in \mathcal{P}$, define $f_P : [0, |P|]_{\mathbb{Z}} \to \mathbb{R}$ such that

$$f_P(m) = \max\{V_P(B; \mathbf{s}|_P, R_P) : B \subset P \text{ and } |B| = m\},$$

which is the maximal profit under the exact constraint of m within group P. So $V_P(A \cap P; \mathbf{s}|_P, R_P) = f_P(|A \cap P|)$ for each $P \in \mathcal{P}$. Define $W : \tau(2^D) \to \mathbb{R}$ such that $W(\mathbf{z}) = \sum_{P \in \mathcal{P}} f_P(\mathbf{z}(P))$ for each $\mathbf{z} \in \tau(2^D)$.

[8] Here, $\mathbf{s}|_P$ is the restriction of salary schedule $\mathbf{s}$ to P.

By Lemma 20 of KSY, each f_P is concave-extensible, and thus W is V-additively separable. Define $U : \tau(2^D) \to \mathbb{R}$ to be associated with T. According to Lemma OA.10, $W + U$ satisfies the vectorial substitutes condition and thus, by Lemma OA.2, is $\mathrm{M}^\natural$-concave; it satisfies a multi-unit single-improvement property (Murota, 2003; Milgrom and Strulovici, 2009). In other words, at least one of the following 3 statements must be true: there exists $\tilde{P} \in \mathcal{P}$ such that $(W+U)(\tau(A) + \mathbf{i}^{\tilde{P}}) > (W+U)(\tau(A))$; there exists $\tilde{P} \in \mathcal{P}$ such that $(W+U)(\tau(A) - \mathbf{i}^{\tilde{P}}) > (W+U)(\tau(A))$; there exists $\tilde{P}, \tilde{P}' \in \mathcal{P}$ such that $(W+U)(\tau(A) - \mathbf{i}^{\tilde{P}} + \mathbf{i}^{\tilde{P}'}) > (W+U)(\tau(A))$.

In the first case, we can use Lemma 19 of KSY to find $d^* \in \tilde{P} \setminus A$ such that $V_{\tilde{P}}((A \cap \tilde{P}) + d^*; \mathbf{s}|_{\tilde{P}}, R_{\tilde{P}}) = f_{\tilde{P}}(|A \cap \tilde{P}| + 1)$. Consequently,

$$V(A + d^*; \mathbf{s}, R+T) = (W+U)(\tau(A) + \mathbf{i}^{\tilde{P}}) > (W+U)(\tau(A)) = V(A; \mathbf{s}, R+T),$$

so $A + d^* \subset D$ is a single-improvement opportunity and we can set $A' = A + d^*$. Similarly, in the second case, we can find $A - d^* \subset D$ with $d^* \in \tilde{P} \cap A$ as a single-improvement opportunity; in the third case, we can find $A - d^* + d^\dagger \subset D$ with $d^* \in \tilde{P} \cap A$ and $d^\dagger \in \tilde{P}' \setminus A$ as a single-improvement opportunity. □

Proof of Theorem 3. The necessity part follows from Proposition 12 because all involved spline concave revenue functions are group concave. For sufficiency, consider a group concave revenue function R, a group separable transfer function T^1, and a cardinally concave transfer function T^2. Lemma OA.12 says that $R + T^1$ satisfies the substitutes condition; and so Theorem 1 implies that $(R + T^1) + T^2$ satisfies the substitutes condition. □

Proof of Theorem 2. The necessity part follows from Proposition 11 because all within-group binary unit-demand revenue functions are group separable. For sufficiency, consider a group separable revenue function R, an additively separable transfer function T^1, and a group concave transfer function T^2. Lemma OA.12 says that $R + T^2$ satisfies the substitutes condition; and so Theorem 1 implies that $(R + T^2) + T^1$ satisfies the substitutes condition. □

OA.2.6 Proofs of Propositions 2 and 14

When $|\mathcal{P}| = 2$, we have the following result.

Lemma OA.13. *Suppose there are two groups. The sum of two vectorial functions that both satisfy the vectorial substitutes condition satisfies it.*

Proof. Given two vectorial functions U and U' that both satisfy the vectorial substitutes condition, we show that $U+U'$ is $\mathrm{M}^\natural$-concave. Consider any $\mathbf{z}, \mathbf{z}' \in \tau(2^D)$ and $P \in \mathcal{P}$ such that $\mathbf{z}(P) > \mathbf{z}'(P)$. There is only one other $P' \in \mathcal{P}$.

If $|\mathbf{z}| \leq |\mathbf{z}'|$, we can apply the second condition in Theorem OA.1 to both U and U' to obtain $U(\mathbf{z}) + U(\mathbf{z}') \leq U(\mathbf{z} - \mathbf{i}^P + \mathbf{i}^{P'}) + U(\mathbf{z}' + \mathbf{i}^P - \mathbf{i}^{P'})$ and $U'(\mathbf{z})+U'(\mathbf{z}') \leq U'(\mathbf{z}-\mathbf{i}^P+\mathbf{i}^{P'})+U'(\mathbf{z}'+\mathbf{i}^P-\mathbf{i}^{P'})$. Combining these two inequalities gives us

$$(U + U')(\mathbf{z}) + (U + U')(\mathbf{z}') \leq (U + U')(\mathbf{z} - \mathbf{i}^P + \mathbf{i}^{P'}) + (U + U')(\mathbf{z}' + \mathbf{i}^P - \mathbf{i}^{P'}).$$

If $|\mathbf{z}| > |\mathbf{z}'|$ and $\mathbf{z}'(P') \geq \mathbf{z}(P')$, we can apply the first condition in Theorem OA.1 to both U and U' to obtain $U(\mathbf{z}) + U(\mathbf{z}') \leq U(\mathbf{z} - \mathbf{i}^P) + U(\mathbf{z}' + \mathbf{i}^P)$ and $U'(\mathbf{z}) + U'(\mathbf{z}') \leq U'(\mathbf{z} - \mathbf{i}^P) + U'(\mathbf{z}' + \mathbf{i}^P)$. (Note that, due to $\mathbf{z}'(P') \geq \mathbf{z}(P')$, we cannot replace $\mathbf{i}^P$ with $\mathbf{i}^{P'}$ in these inequalities.) Combining these two inequalities gives us

$$(U + U')(\mathbf{z}) + (U + U')(\mathbf{z}') \leq (U + U')(\mathbf{z} - \mathbf{i}^P) + (U + U')(\mathbf{z}' + \mathbf{i}^P).$$

If $|\mathbf{z}| > |\mathbf{z}'|$ and $\mathbf{z}'(P') < \mathbf{z}(P')$, we can apply the definition of $\mathrm{M}^\natural$-concavity to U and U'. Because the second case of the definition is never possible due to $\mathbf{z}'(P') < \mathbf{z}(P')$, the first case must be true. So $U(\mathbf{z})+U(\mathbf{z}') \leq U(\mathbf{z}-\mathbf{i}^P)+U(\mathbf{z}'+\mathbf{i}^P)$ and $U'(\mathbf{z})+U'(\mathbf{z}') \leq U'(\mathbf{z}-\mathbf{i}^P)+U'(\mathbf{z}'+\mathbf{i}^P)$. They are the same as the last case. □

Proof of Proposition 14. For necessity, since the zero vectorial function satisfies the vectorial substitutes condition, always preserving it requires a vectorial function to satisfy it itself. The sufficiency part follows from Lemma OA.13. □

Proof of Proposition 2. We start with a group concave revenue function R, a group separable transfer function T, and a group concave transfer function T'. To show that $R+(T+T')$ satisfies the substitutes condition, we first note that $R+T'$ satisfies it by Lemma OA.13. Then, we know $(R + T') + T$ satisfies it by Theorem 2. □

OA.2.7 Proof of Proposition 3

Proof of Proposition 3. For the first statement, to show $X(\,\cdot\,; R + \Upsilon)$ satisfies the substitutes condition, we start with any two salary schedules $\bar{\mathbf{s}}$ and $\bar{\mathbf{s}}'$ with $\bar{\mathbf{s}}' \geq \bar{\mathbf{s}}$, and any $A^* \in X(\bar{\mathbf{s}}; R + \Upsilon)$.

Define two new salary schedules $\hat{\mathbf{s}}$ and $\hat{\mathbf{s}}'$ such that $\hat{s}_d = \tilde{g}^d(\bar{s}_d)$ and $\hat{s}'_d = \tilde{g}^d(\bar{s}'_d)$ for every $d \in D$. Since every $\tilde{g}^d$ is strictly increasing, we have $\hat{\mathbf{s}}' \geq \hat{\mathbf{s}}$. We also have,

for every $A \subset D$, $V(A; \bar{\mathbf{s}}, R + \Upsilon) = V(A; \hat{\mathbf{s}}, R)$ and $V(A; \bar{\mathbf{s}}', R + \Upsilon) = V(A; \hat{\mathbf{s}}', R)$; and thus $X(\bar{\mathbf{s}}; R + \Upsilon) = X(\hat{\mathbf{s}}; R)$ and $X(\bar{\mathbf{s}}'; R + \Upsilon) = X(\hat{\mathbf{s}}'; R)$.

Since $X(\,\cdot\,; R)$ satisfies the substitutes condition, given $A^* \in X(\bar{\mathbf{s}}; R + \Upsilon) = X(\hat{\mathbf{s}}; R)$, there exists $A' \in X(\hat{\mathbf{s}}'; R) = X(\bar{\mathbf{s}}'; R + \Upsilon)$ such that

$$\{d \in A^* : \bar{s}_d = \bar{s}'_d\} = \{d \in A^* : \hat{s}_d = \hat{s}'_d\} \subset A'.$$

Thus, $X(\,\cdot\,; R + \Upsilon)$ satisfies the substitutes condition.

For the second statement, to show $X(\,\cdot\,; R)$ satisfies the substitutes condition, we start with any two salary schedules $\hat{\mathbf{s}}$ and $\hat{\mathbf{s}}'$ with $\hat{\mathbf{s}}' \geq \hat{\mathbf{s}}$, and any $A^* \in X(\hat{\mathbf{s}}; R)$.

Define two new salary schedules $\bar{\mathbf{s}}$ and $\bar{\mathbf{s}}'$ such that $\hat{s}_d = \tilde{g}^d(\bar{s}_d)$ and $\hat{s}'_d = \tilde{g}^d(\bar{s}'_d)$ for every $d \in D$. Since every $\tilde{g}^d(s_d)$ is strictly increasing and $\tilde{g}^d(\mathbb{R}) = \mathbb{R}$, we know $\bar{\mathbf{s}}$ and $\bar{\mathbf{s}}'$ are well-defined, and $\bar{\mathbf{s}}' \geq \bar{\mathbf{s}}$. We also have $X(\bar{\mathbf{s}}; R + \Upsilon) = X(\hat{\mathbf{s}}; R)$ and $X(\bar{\mathbf{s}}'; R + \Upsilon) = X(\hat{\mathbf{s}}'; R)$.

Since $X(\,\cdot\,; R + \Upsilon)$ satisfies the substitutes condition, given $A^* \in X(\hat{\mathbf{s}}; R) = X(\bar{\mathbf{s}}; R + \Upsilon)$, there exists $A' \in X(\bar{\mathbf{s}}'; R + \Upsilon) = X(\hat{\mathbf{s}}'; R)$ such that

$$\{d \in A^* : \hat{s}_d = \hat{s}'_d\} = \{d \in A^* : \bar{s}_d = \bar{s}'_d\} \subset A'.$$

Thus, $X(\,\cdot\,; R)$ satisfies the substitutes condition. □

OA.2.8 Proof of Proposition 4

Lemma OA.14 provides a novel characterization of the substitutes condition. The new condition is interesting in that it appears to be weaker than the no complementarities condition of Gul and Stacchetti (1999), so the sufficiency part of the lemma becomes handy in the proof of Proposition 4. For each $d \in D$, define $\mathbf{e}^d$ as the salary schedule with $\mathbf{e}^d(d) = 1$ and $\mathbf{e}^d(d') = 0$ for each $d' \in (D - d)$.

Lemma OA.14. *A revenue function R satisfies the substitutes condition if and only if for each salary schedule $\mathbf{s}$, all demand sets $A, B \in X(\mathbf{s}; R)$, and each worker $d \in A \setminus B$, there exists a demand set $C \in X(\mathbf{s}; R)$ with $A \cap C = A - d$.*

Proof. For necessity, observe from $B \in X(\mathbf{s}; R)$ and $d \notin B$ that $B \in X(\mathbf{s} + \mathbf{e}^d; R)$. By the substitutes condition, $A \in X(\mathbf{s}; R)$ implies the existence of $C \in X(\mathbf{s} + \mathbf{e}^d; R)$ with $A - d \subset C$. We know

$$V(B; \mathbf{s}, R) = V(B; \mathbf{s} + \mathbf{e}^d, R) = V(C; \mathbf{s} + \mathbf{e}^d, R) \leq V(C; \mathbf{s}, R) \leq V(B; \mathbf{s}, R),$$

where the last inequality follows from $B \in X(\mathbf{s}; R)$. So all items above takes the same value. We thus have $C \in X(\mathbf{s}; R)$ and $d \notin C$, so $A \cap C = A - d$.

For sufficiency, to show that R satisfies the substitutes condition, we only need to establish that for all salary schedule $\mathbf{s}^*$, $A^* \in X(\mathbf{s}^*; R)$, $d^* \in A^*$, and real number $\delta^* > 0$, there exists $C^* \in X(\mathbf{s}^* + \delta^* \mathbf{e}^{d^*}; R)$ such that $A^* - d^* \subset C^*$. In one case, there exists $C \in X(\mathbf{s}^* + \delta^* \mathbf{e}^{d^*}; R)$ which contains d^*. Then $A^* \in X(\mathbf{s}^* + \delta^* \mathbf{e}^{d^*}; R)$, and we are done by setting $C^* = A^*$.

In the other case, no demand set $C \in X(\mathbf{s}^* + \delta^* \mathbf{e}^{d^*}; R)$ contains d^*. Define $\delta' \coloneqq \sup\{\delta \in \mathbb{R} : A^* \in X(\mathbf{s}^* + \delta \mathbf{e}^{d^*}; R)\}$, and let $\hat{\mathbf{s}} \coloneqq \mathbf{s}^* + \delta' \mathbf{e}^{d^*}$. It is easy to see $\delta' \in [0, \delta^*)$, $A^* \in X(\hat{\mathbf{s}}; R)$, $A^* \notin X(\mathbf{s}^* + \delta'' \mathbf{e}^{d^*}; R)$ for each $\delta'' > \delta'$, and the existence of $B^* \in X(\hat{\mathbf{s}}; R)$ with $d^* \notin B^*$. So given $A^*, B^* \in X(\hat{\mathbf{s}}; R)$ and $d^* \in A^* \setminus B^*$, by assumption, there exists $C^* \in X(\hat{\mathbf{s}}; R)$ with $A^* \cap C^* = A^* - d^*$.

For each $B \in X(\mathbf{s}^* + \delta^* \mathbf{e}^{d^*}; R)$, we have

$$V(B; \hat{\mathbf{s}}, R) \leq V(C^*; \hat{\mathbf{s}}, R) = V(C^*; \mathbf{s}^* + \delta^* \mathbf{e}^{d^*}, R) \leq V(B; \mathbf{s}^* + \delta^* \mathbf{e}^{d^*}, R) = V(B; \hat{\mathbf{s}}, R).$$

So all items above takes the same value, and, accordingly, $C^* \in X(\mathbf{s}^* + \delta^* \mathbf{e}^{d^*}; R)$, and we are done. □

Proof of Proposition 4. Take a complex transfer function Υ that preserves the substitutes condition for a class of revenue functions $\mathcal{R}$ closed under translation. Given an arbitrary $\bar{\mathbf{s}} \in \mathbb{R}^D$, define $\bar{T} \coloneqq \Upsilon(\,\cdot\,, \bar{\mathbf{s}})$. We only need to prove that, given any $R \in \mathcal{R}$, $R + \bar{T}$ satisfies the substitutes condition. To utilize the sufficiency part of Lemma OA.14, we begin with a salary schedule $\hat{\mathbf{s}}$, demand sets $A^*, B^* \in X(\hat{\mathbf{s}}; R + \bar{T})$, and a worker $d^* \in A^* \setminus B^*$, and show that their existence implies the existence of a demand set $B^\diamond \in X(\hat{\mathbf{s}}; R + \bar{T})$ with $A^* \cap B^\diamond = A^* - d^*$.

Define a new revenue function R' such that $R'(B) = R(B) + \sum_{d \in B} (\bar{s}_d - \hat{s}_d)$ for each $B \subset D$. We have $R' \in \mathcal{R}$, and for each $B \subset D$,

$$\begin{aligned} V(B; \bar{\mathbf{s}}, R' + \Upsilon) &= R'(B) + \Upsilon(B, \bar{\mathbf{s}}) - \sum_{d \in B} \bar{s}_d \\ &= R(B) + \sum_{d \in B} (\bar{s}_d - \hat{s}_d) + \bar{T}(B) - \sum_{d \in B} \bar{s}_d \\ &= R(B) + \bar{T}(B) - \sum_{d \in B} \hat{s}_d \\ &= V(B; \hat{\mathbf{s}}, R + \bar{T}). \end{aligned}$$

This implies $X(\bar{\mathbf{s}}; R' + \Upsilon) = X(\hat{\mathbf{s}}; R + \bar{T})$. Specifically, $A^*, B^* \in X(\hat{\mathbf{s}}; R + \bar{T}) = X(\bar{\mathbf{s}}; R' + \Upsilon)$.

Pick an arbitrary $\delta > 0$. By the two requirements for complex transfer functions,

every $B \in X(\hat{\mathbf{s}}; R + \bar{T})$ with $d^* \notin B$, including B^*, belongs to $X(\bar{\mathbf{s}} + \delta \mathbf{e}^{d^*}; R' + \Upsilon)$, while no other $B \subset D$ does. In other words,

$$X(\bar{\mathbf{s}} + \delta \mathbf{e}^{d^*}; R' + \Upsilon) = \{B' \in X(\hat{\mathbf{s}}; R + \bar{T}) \,:\, d^* \notin B'\}. \tag{OA.7}$$

Since Υ preserves the substitutes condition for R', we know $X(\,\cdot\,; R' + \Upsilon)$ satisfies the substitutes condition. Hence, for $A^* \in X(\bar{\mathbf{s}}; R' + \Upsilon)$, there exists $B^\diamond \in X(\bar{\mathbf{s}} + \delta \mathbf{e}^{d^*}; R' + T)$ such that $A^* - d^* \subset B^\diamond$. By Equation (OA.7), $B^\diamond \in X(\hat{\mathbf{s}}; R + \bar{T})$ with $d^* \notin B^\diamond$. So $A^* \cap B^\diamond = A^* - d^*$. □

OA.2.9 Proofs of Corollary 2 and Proposition 5

Proof of Corollary 2. Given a C-additively separable complex transfer function Υ, a group concave transfer function T, and a group separable revenue function R, we know, by Theorem 2, that $X(\,\cdot\,; R + T)$ satisfies the substitutes condition. By the first part of Proposition 3, $X(\,\cdot\,; (R + T) + \Upsilon)$ satisfies the substitutes condition. □

Lemma OA.15. *If for a complex transfer function Υ, given any fixed salary schedule $\mathbf{s}$, the transfer function $T^{\mathbf{s}} \coloneqq \Upsilon(\,\cdot\,, \mathbf{s})$ preserves the substitutes condition for all group separable revenue functions, then Υ is the sum of a C-additively separable complex transfer function and a C-group concave function.*

Proof. By the proof of Proposition 11, for each $\mathbf{s} \in \mathbb{R}^D$, we can define an additively separable function $\bar{T}^{\mathbf{s}}$ such that $\bar{T}^{\mathbf{s}}(A) = \sum_{d \in A} T^{\mathbf{s}}(\{d\}) = \sum_{d \in A} \Upsilon(\{d\}, \mathbf{s})$ for each $A \subset D$, and obtain a group concave transfer function $\hat{T}^{\mathbf{s}} \coloneqq T^{\mathbf{s}} - \bar{T}^{\mathbf{s}}$. In other words, for all $A \subset D$ and $\mathbf{s} \in \mathbb{R}^D$,

$$\Upsilon(A, \mathbf{s}) = T^{\mathbf{s}}(A) = \bar{T}^{\mathbf{s}}(A) + \hat{T}^{\mathbf{s}}(A) = \sum_{d \in A} \Upsilon(\{d\}, \mathbf{s}) + \hat{T}^{\mathbf{s}}(A).$$

Since Υ is a complex transfer function, the term $\sum_{d \in A} \Upsilon(\{d\}, \mathbf{s})$ as a function of $A \subset D$ is a C-additively separable complex transfer function. We shall prove that the term $\hat{T}^{\mathbf{s}}(A)$ as a complex function of $A \subset D$ and $\mathbf{s} \in \mathbb{R}^D$ is C-group concave. The first requirement for C-group concavity is already established. For the second, pick any $\bar{\mathbf{s}}, \bar{\mathbf{s}}' \in \mathbb{R}^D$ and any $A \subset D$ such that $\bar{\mathbf{s}}'|_P = \bar{\mathbf{s}}|_P$ for every group $P \subset A$. We now show $\hat{T}^{\bar{\mathbf{s}}'}(A) = \hat{T}^{\bar{\mathbf{s}}}(A)$.

Because we can turn $\bar{\mathbf{s}}'$ into $\bar{\mathbf{s}}$ by replacing $\bar{s}'_d$ with $\bar{s}_d$ one worker d at a time, we can assume without loss of generality that $\bar{s}'_{d^*} \neq \bar{s}_{d^*}$ only for one worker d^*. By assumption, either $d^* \notin A$, or there exists $d' \neq d^*$ such that $d^* \in A$, $d' \notin A$, and $d^*, d' \in P^*$ for a group P^*. In the former case, since $\hat{T}^{\mathbf{s}}(A) = \Upsilon(A, \mathbf{s}) - \sum_{d \in A} \Upsilon(\{d\}, \mathbf{s})$ is independent of any s_d with $d \notin A$, we know

$\hat{T}^{\bar{\mathbf{s}}'}(A) = \hat{T}^{\bar{\mathbf{s}}}(A)$ indeed holds. In the latter case, $\hat{T}^{\bar{\mathbf{s}}'}(A) = \hat{T}^{\bar{\mathbf{s}}'}(A - d^* + d') = \hat{T}^{\bar{\mathbf{s}}}(A - d^* + d') = \hat{T}^{\bar{\mathbf{s}}}(A)$, where the first and last equalities follow from group concavity of $\hat{T}^{\bar{\mathbf{s}}'}$ and $\hat{T}^{\bar{\mathbf{s}}}$, and the second equality follows from $d^* \notin A - d^* + d'$. □

Proof of Proposition 5. Start with a complex transfer function Υ that preserves the substitutes condition for all group separable revenue functions. Since the class of group separable revenue functions is closed under translation, we know from Proposition 4 that given any fixed salary schedule $\mathbf{s}$, $T^{\mathbf{s}} \coloneqq \Upsilon(\,\cdot\,, \mathbf{s})$, as a transfer function, preserves the substitutes condition for all group separable revenue functions. The conclusion follows from Lemma OA.15. □

OA.2.10 Proof of Theorem 4

Proof of Theorem 4. For necessity, let a complex transfer function Υ always preserves the substitutes condition. Let there be only one group. Then, the class of group separable revenue functions is the same as the class of revenue functions satisfying the substitutes condition. By the proof of Proposition 5, we know for all $A \subset D$ and $\mathbf{s} \in \mathbb{R}^D$, $\Upsilon(A, \mathbf{s}) = \sum_{d \in A} \Upsilon(\{d\}, \mathbf{s}) + \Upsilon'(A, \mathbf{s})$, where Υ' is a C-group concave complex function. Because there is only one group, by the definition of C-group concavity, we know for all $A \subsetneq D$ and $\mathbf{s} \in \mathbb{R}^D$, $\Upsilon'(A, \mathbf{s}) = \bar{T}(A)$, where $\bar{T}$ is a group concave function and thus cardinally concave; also, for each $\mathbf{s} \in \mathbb{R}^D$, $T^{\mathbf{s}} \coloneqq \Upsilon'(\,\cdot\,, \mathbf{s})$ is cardinally concave.

By cardinal concavity of $T^{\mathbf{s}}$, we know

$$T^{\mathbf{s}}(D) - T^{\mathbf{s}}(D - d) \leq T^{\mathbf{s}}(D - d) - T^{\mathbf{s}}(D - d - d')$$

for all distinct $d, d' \in D$, so

$$T^{\mathbf{s}}(D) \leq 2T^{\mathbf{s}}(D - d) - T^{\mathbf{s}}(D - d - d') = 2\bar{T}(D - d) - \bar{T}(D - d - d') < \infty.$$

Accordingly, we can define a transfer function T such that $T(D) = \sup\{T^{\mathbf{s}}(D) : \mathbf{s} \in \mathbb{R}^D\}$ and $T(A) = \bar{T}(A)$ for every $A \subsetneq D$. It is simple to check that T is cardinally concave. Also, $\Upsilon(D, \mathbf{s}) \leq T(D)$ for each $\mathbf{s} \in \mathbb{R}^D$. So the only statement left for us to prove is the weakly increasing property of $\Upsilon'(D, \mathbf{s})$ as a function of $\mathbf{s} \in \mathbb{R}^D$.

Define $F : \mathbb{R}^D \to (-\infty, 0]$ such that $F(\mathbf{s}) = T^{\mathbf{s}}(D) - T(D)$ for every $\mathbf{s} \in \mathbb{R}^D$. We need to show F is weakly increasing. Suppose otherwise: there exists $\mathbf{s}^* \in \mathbb{R}^D$, $d^* \in D$ and $\delta > 0$ such that $F(\mathbf{s}^* + \delta \mathbf{e}^{d^*}) < F(\mathbf{s}^*)$. Pick another worker $d' \neq d^*$,[9] and a small $\epsilon > 0$ such that $\epsilon < \min\{F(\mathbf{s}^*) - F(\mathbf{s}^* + \delta \mathbf{e}^{d^*}), 1\}$.

[9] Here, we implicitly assume that there are at least two workers in the economy. For the $M = 1$ case, the theorem is trivially true.

For notational simplicity, for each $d \in D$, define $g^d : \mathbb{R} \to \mathbb{R}$ such that $g^d(s_d) = \Upsilon(\{d\}, \mathbf{s})$ for each $\mathbf{s} \in \mathbb{R}^D$. Consider an additively separable revenue function R satisfying

$$\begin{aligned}
R(\{d^*\}) &= s^*_{d^*} + \delta - g^{d^*}(s^*_{d^*} + \delta) + T(D - d^*) - T(D) - F(\mathbf{s}^* + \delta \mathbf{e}^{d^*}) + 1; \\
R(\{d'\}) &= s^*_{d'} - g^{d'}(s^*_{d'}) + T(D - d') - T(D) - F(\mathbf{s}^*) + \epsilon; \\
R(\{d\}) &= s^*_{d} - g^{d}(s^*_{d}) + T(D - d) - T(D) - F(\mathbf{s}^* + \delta \mathbf{e}^{d^*}) + 1,
\end{aligned}$$

for $d \in D \setminus \{d^*, d'\}$.

Accordingly, we have

$$V(D; \mathbf{s}^*, R + \Upsilon) - V(D - d'; \mathbf{s}^*, R + \Upsilon) = \epsilon > 0,$$

and

$$V(D; \mathbf{s}^* + \delta \mathbf{e}^{d^*}, R + \Upsilon) - V(D - d'; \mathbf{s}^* + \delta \mathbf{e}^{d^*}, R + \Upsilon) = F(\mathbf{s}^* + \delta \mathbf{e}^{d^*}) - F(\mathbf{s}^*) + \epsilon < 0.$$

For each $d \in D \setminus \{d^*, d'\}$, we have

$$V(D; \mathbf{s}^*, R + \Upsilon) - V(D - d; \mathbf{s}^*, R + \Upsilon) = [F(\mathbf{s}^*) - F(\mathbf{s}^* + \delta \mathbf{e}^{d^*})] + 1 > 1 > 0,$$

and

$$V(D; \mathbf{s}^* + \delta \mathbf{e}^{d^*}, R + \Upsilon) - V(D - d; \mathbf{s}^* + \delta \mathbf{e}^{d^*}, R + \Upsilon) = 1 > 0.$$

By the second requirement for Υ to be a complex transfer function, we have $s^*_{d^*} + \delta - g^{d^*}(s^*_{d^*} + \delta) > s^*_{d^*} - g^{d^*}(s^*_{d^*})$. So

$$\begin{aligned}
&V(D; \mathbf{s}^*, R + \Upsilon) - V(D - d^*; \mathbf{s}^*, R + \Upsilon) \\
&\qquad = [\delta - g^{d^*}(s^*_{d^*} + \delta) + g^{d^*}(s^*_{d^*})] + [F(\mathbf{s}^*) - F(\mathbf{s}^* + \delta \mathbf{e}^{d^*})] + 1 > 1 > 0,
\end{aligned}$$

and

$$V(D; \mathbf{s}^* + \delta \mathbf{e}^{d^*}, R + \Upsilon) - V(D - d^*; \mathbf{s}^* + \delta \mathbf{e}^{d^*}, R + \Upsilon) = 1 > 0.$$

For all $A \subsetneq D$, $d \in A$, and $\mathbf{s} \in \mathbb{R}^D$, we have

$$\begin{aligned}
&[V(A; \mathbf{s}, R + \Upsilon) - V(A - d; \mathbf{s}, R + \Upsilon)] - [V(D; \mathbf{s}, R + \Upsilon) - V(D - d; \mathbf{s}, R + \Upsilon)] \\
&\qquad = [T^{\mathbf{s}}(A) - T^{\mathbf{s}}(A - d)] - [T^{\mathbf{s}}(D) - T^{\mathbf{s}}(D - d)] \geq 0,
\end{aligned}$$

where the inequality follows from the cardinal concavity of $T^{\mathbf{s}}$. So we can combine all the above inequalities to derive that no $A \subset D$ with $|D \setminus A| > 1$ can be a demand set at salary schedules $\mathbf{s}^*$ or $\mathbf{s}^* + \delta \mathbf{e}^{d^*}$. We thus have $X(\mathbf{s}^*; R + \Upsilon) = \{D\}$ and

$X(\mathbf{s}^* + \delta \mathbf{e}^{d^*}; R + \Upsilon) = \{D - d'\}$. This contradicts the assumption that Υ preserves the substitutes condition for R.

For sufficiency, choose any revenue function R satisfying the substitutes condition, any C-additively separable complex transfer function Υ associated with individual transfer functions $\{g^d\}_{d \in D}$, and any C-cardinally concave complex function Υ' associated with a cardinally concave function T such that for each $\mathbf{s} \in \mathbb{R}^D$, the transfer function $T^{\mathbf{s}} \coloneqq \Upsilon'(\,\cdot\,, \mathbf{s})$ is cardinally concave. Further assume $\Upsilon + \Upsilon'$ is a complex transfer function. To show $X(\,\cdot\,; R + \Upsilon + \Upsilon')$ satisfies the substitutes condition, we start with two arbitrary salary schedules $\bar{\mathbf{s}}$ and $\bar{\mathbf{s}}'$ with $\bar{\mathbf{s}}' \geq \bar{\mathbf{s}}$, and any $A^* \in X(\bar{\mathbf{s}}; R + \Upsilon + \Upsilon')$.

Define two salary schedules $\hat{\mathbf{s}}$ and $\hat{\mathbf{s}}'$ such that $\hat{s}_d = \bar{s}_d - g^d(\bar{s}_d)$ and $\hat{s}'_d = \bar{s}'_d - g^d(\bar{s}'_d)$ for every $d \in D$. Since Υ is a complex transfer function, we have $\hat{\mathbf{s}}' \geq \hat{\mathbf{s}}$. It is routine to check that for all $A \subset D$ and $\mathbf{s} \in \mathbb{R}^D$, $V(A; \bar{\mathbf{s}}, R + \Upsilon + \Upsilon') = V(A; \hat{\mathbf{s}}, R + T^{\bar{\mathbf{s}}})$ and $V(A; \bar{\mathbf{s}}', R + \Upsilon + \Upsilon') = V(A; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}'})$. Thus, we have $X(\bar{\mathbf{s}}; R + \Upsilon + \Upsilon') = X(\hat{\mathbf{s}}; R + T^{\bar{\mathbf{s}}})$ and $X(\bar{\mathbf{s}}'; R + \Upsilon + \Upsilon') = X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}'})$.

In one case, $D \in X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}'}) = X(\bar{\mathbf{s}}'; R + \Upsilon + \Upsilon')$. We have

$$\{d \in A^* : \bar{s}_d = \bar{s}'_d\} = \{d \in A^* : \hat{s}_d = \hat{s}'_d\} \subset D \in X(\bar{\mathbf{s}}'; R + \Upsilon + \Upsilon').$$

In the other case, $D \notin X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}'})$. In other words, there exists $A^\diamond \subsetneq D$ such that $V(A^\diamond; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}'}) > V(D; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}'})$. Since Υ' is C-cardinally concave, $T^{\bar{\mathbf{s}}}(A) = T^{\bar{\mathbf{s}}'}(A)$ for every $A \subsetneq D$ and $T^{\bar{\mathbf{s}}}(D) \leq T^{\bar{\mathbf{s}}'}(D)$. So

$$V(A^\diamond; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}}) = V(A^\diamond; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}'}) > V(D; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}'}) \geq V(D; \hat{\mathbf{s}}', R + T^{\bar{\mathbf{s}}}),$$

which implies $D \notin X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}})$. Hence, $X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}}) = X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}'})$.

According to the sufficiency part of Theorem 2, $X(\,\cdot\,; R + T^{\bar{\mathbf{s}}})$ satisfies the substitutes condition. Therefore, for the given $A^* \in X(\bar{\mathbf{s}}; R + \Upsilon + \Upsilon') = X(\hat{\mathbf{s}}; R + T^{\bar{\mathbf{s}}})$, there exists $A' \in X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}}) = X(\hat{\mathbf{s}}'; R + T^{\bar{\mathbf{s}}'}) = X(\bar{\mathbf{s}}'; R + \Upsilon + \Upsilon')$ such that

$$\{d \in A^* : \bar{s}_d = \bar{s}'_d\} = \{d \in A^* : \hat{s}_d = \hat{s}'_d\} \subset A' \in X(\bar{\mathbf{s}}'; R + \Upsilon + \Upsilon').$$

In both cases, the requirement for $X(\,\cdot\,; R + \Upsilon + \Upsilon')$ to satisfy the substitutes condition is fulfilled. □

OA.2.11 Proof of Theorem 8

We first prove a result analogous to Lemma 1 of Yang (2017).

Lemma OA.16. *If for a complex transfer function Υ, given some fixed salary schedule $\mathbf{s}^*$, the transfer function $T^{\mathbf{s}^*} \coloneqq \Upsilon(\,\cdot\,; \mathbf{s}^*)$ fails to always preserve*

the substitutes condition, then there exists a revenue function R^* satisfying the substitutes condition, and $A, B \in X(\mathbf{s}^*; R^* + \Upsilon)$ such that $|A \setminus B| \geq 2$ and $A \subset C \subset A \cup B$ for every $C \in X(\mathbf{s}^*; R^* + \Upsilon) \setminus \{B\}$.

Proof. Consider a revenue function R such that it satisfies the substitutes condition but $R' \coloneqq R + T^{\mathbf{s}^*}$ does not. Then, by Theorem 3 and Lemma 1 of Yang (2017), there exists $\mathbf{s}' \in \mathbb{R}^D$, and $A, B \in X(\mathbf{s}'; R')$ such that $|A \setminus B| \geq 2$ and $A \subset C \subset A \cup B$ for every $C \in X(\mathbf{s}'; R') \setminus \{B\}$.

Define a revenue function R^* such that $R^*(C) = R(C) + \sum_{d \in C}(s^*_d - s'_d)$ for every $C \subset D$. By Proposition 1, R^* satisfies the substitutes condition. We have $V(C; \mathbf{s}^*, R^* + \Upsilon) = V(C; \mathbf{s}', R')$ for every $C \subset D$, and thus $X(\mathbf{s}^*; R^* + \Upsilon) = X(\mathbf{s}'; R')$. Therefore, $A, B \in X(\mathbf{s}^*; R^* + \Upsilon)$ and the conclusion holds. □

Another lemma disciplines the requirement of respecting the law of maintained demand.

Lemma OA.17. *If a complex transfer function Υ respects the law of maintained demand but fails to always preserve the substitutes condition, then there exists a salary schedule $\mathbf{s}$ such that the transfer function $T^{\mathbf{s}} \coloneqq \Upsilon(\cdot; \mathbf{s})$ fails to always preserve the substitutes condition.*

Proof. Suppose given any fixed salary schedule $\mathbf{s}$, the transfer function $T^{\mathbf{s}} \coloneqq \Upsilon(\cdot; \mathbf{s})$ always preserves the substitutes condition. Apply Lemma OA.15 to the case of there existing only one group, we know that there exists a cardinally concave transfer function $\bar{T}$ and a function $F : \mathbb{R}^D \to \mathbb{R}$ such that for all $\mathbf{s} \in \mathbb{R}^D$ and $A \subsetneq D$, $\Upsilon(A, \mathbf{s}) = \sum_{d \in A} \Upsilon(\{d\}, \mathbf{s}) + \bar{T}(A)$, and $\Upsilon(D, \mathbf{s}) = \sum_{d \in D} \Upsilon(\{d\}, \mathbf{s}) + \bar{T}(D) + F(\mathbf{s})$. By the argument in the proof of the necessity part of Theorem 4 (second paragraph), we know $F(\mathbf{s}) \leq 0$ for every $\mathbf{s} \in \mathbb{R}^D$.

Since Υ fails to always preserve the substitutes condition, by the sufficiency part of Theorem 4, F is not weakly increasing. There exists $\mathbf{s}^* \in \mathbb{R}^D$, $d^* \in D$, and $\delta > 0$ such that $F(\mathbf{s}^* + \delta \mathbf{e}^{d^*}) < F(\mathbf{s}^*)$. We can define an additively separable revenue function $R : 2^D \to \mathbb{R}$ as in the proof of necessity part of Theorem 4, and similarly derive $X(\mathbf{s}^*; R + \Upsilon) = \{D\}$ and $X(\mathbf{s}^* + \delta \mathbf{e}^{d^*}; R + \Upsilon) = \{D - d'\}$. This is a violation of the law of maintained demand for $X(\mathbf{s}; R + \Upsilon)$. □

Proof of Theorem 8. Consider a proper complex transfer function Υ which respects the law of maintained demand but fails to always preserve the substitutes condition.

By Lemma OA.17, there exists $\mathbf{s}^* \in \mathbb{R}^D$ such that the transfer function $T^{\mathbf{s}^*} \coloneqq \Upsilon(\cdot\,; \mathbf{s}^*)$ fails to always preserve the substitutes condition. Then, by Lemma OA.16, there exists a revenue function R_1 satisfying the substitutes condition, and $A, B \in X(\mathbf{s}^*; R_1 + \Upsilon)$ such that $|A \setminus B| \geq 2$ and $A \subset C \subset A \cup B$ for every $C \in X(\mathbf{s}^*; R_1 + \Upsilon) \setminus \{B\}$.

The marginal revenue from any $d \in D$ given R_1 is bounded above, say, by $\Xi > 0$. So by the second requirement for Υ to be proper, there exists $\hat{\Xi} \coloneqq \max\{\Delta_d : d \in D\} > 0$ such that for all $\mathbf{s} \in \mathbb{R}^D$, $C \in X(\mathbf{s}; R_1 + \Upsilon)$, and $d \in D$ with $s_d \geq \hat{\Xi}$, we have $d \notin C$. Define $\bar{\mathbf{s}} \in \mathbb{R}^D$ such that

$$\bar{s}_d = \begin{cases} s^*_d, & \text{if } d \in B; \\ s^*_d - \delta, & \text{if } d \in A \setminus B; \\ \hat{\Xi}, & \text{if } d \in D \setminus (A \cup B). \end{cases}$$

Here, δ is a positive number sufficiently small that $V(C; \bar{\mathbf{s}}, R_1 + \Upsilon) < V(B; \mathbf{s}^*, R_1 + \Upsilon) = \Pi(\mathbf{s}^*; R_1 + \Upsilon)$ for every $C \notin X(\mathbf{s}^*; R_1 + \Upsilon)$; this is possible thanks to the first requirement for Υ to be proper. Define a revenue function R_2 such that for every $C \subset D$,

$$R_2(C) = \sum_{d \in C} \bar{s}_d + \Xi' \cdot \min\{1, |C \cap (A \setminus B)|\},$$

where $\Xi' \coloneqq \hat{\Xi} + \sum_{d \in D} |\bar{s}_d|$. Clearly, R_2 satisfies the substitutes condition because it is the sum of an additively separable function and a unit-demand function.

Consider a job matching setting where firms 1 and 2 are endowed with revenue functions R_1 and R_2 respectively, and firm 1 faces Υ. The utility functions of workers are quasilinear in salaries: $U_{d,h}(s) = s$ for all $d \in D$, $h \in H$, and $s \in \mathbb{R}$. The utility of unemployment is $-\infty$.[10] We show that there exists no competitive equilibrium. Suppose, to the contrary, that there exists an equilibrium in which both firms face salary schedule $\hat{\mathbf{s}}$ (this is without loss of generality due to the symmetry in workers' utility setup), and are matched to $A_1, A_2 \subset D$ respectively. We know $A_1 \cup A_2 = D$ because each worker's utility from unemployment is $-\infty$ (while revenue and transfer functions are all finite by assumption).

First, we show $A_1 \subset A \cup B$. Suppose there exists $d^* \in A_1$, but $d^* \in D \setminus (A \cup B)$. By the definition of R_2, the revenue created by d^* for firm 2 is Ξ. Because $d^* \notin A_2$, firm 1 must pay d^* at least Ξ in salary, but this contradicts the definition of Ξ.

[10] This can thus be viewed as the object assignment setting of Gul and Stacchetti (1999) and Yang (2017), where every invisible good is assigned in a competitive equilibrium.

Second, we show $A_2 \cap (A \setminus B) \neq \emptyset$, and thus $A \setminus B \nsubseteq A_1$. Suppose otherwise. Then the revenue created by adding any $d^* \in A \setminus B$ for firm 2 is $\bar{s}_{d^*} + \Xi'$. Because $d^* \notin A_2$, firm 1 must pay d^* at least $\bar{s}_{d^*} + \Xi' > \Xi$ in salary, but this contradicts the definition of Ξ.

Define $\mathbf{s}' \in \mathbb{R}^D$ by

$$s'_d = \begin{cases} s^*_d, & \text{if } d \in B; \\ \hat{s}_d, & \text{otherwise.} \end{cases}$$

We now show that $\mathbf{s}'$ supports the same allocation in a competitive equilibrium. Because we can turn $\hat{\mathbf{s}}$ into $\mathbf{s}'$ one salary at a time, we assume without loss of generality that $s^*_d = \hat{s}_d$ for every $d \in D$ except $d^\diamond \in B$.

In the first case, $d^\diamond \in A_1$. Since $d^\diamond \notin A_2$ and the revenue created by adding $d^\diamond$ to firm 2 is $s^*_{d^\diamond}$, we know $s^*_{d^\diamond} < \hat{s}_{d^\diamond}$. After $\hat{s}_{d^\diamond}$ decreases to $s^*_{d^\diamond}$, A_1 is still a demand set for firm 1 because Υ respects the law of maintained demand; A_2 is still a demand set for firm 2 because it is indifferent between hiring $d^\diamond$ or not. Equilibrium is maintained.

In the second case, $d^\diamond \in A_2$. Since firm 2 is willing to hire $d^\diamond$ at $\hat{\mathbf{s}}$, we know $s^*_{d^\diamond} > \hat{s}_{d^\diamond}$. After $\hat{s}_{d^\diamond}$ increases to $s^*_{d^\diamond}$, A_1 is still a demand set for firm 1 due to the second requirement for Υ to be a complex transfer function; A_2 is still a demand set for firm 2. Equilibrium is also maintained.

We show contradictions case by case.

In the first case, $A_1 \setminus B \neq \emptyset$. For each $d \in A_1 \setminus B$, we have $s'_d \geq R_2(A_2 + d) - R_2(A_2) = \bar{s}_d = s^*_d - \delta$. Since $A \subset C \subset A \cup B$ for every $C \in X(\mathbf{s}^*; R_1 + \Upsilon) \setminus \{B\}$, we know $A_1 \notin X(\mathbf{s}^*; R_1 + \Upsilon)$. Hence,

$$\begin{aligned} V(A_1; \mathbf{s}', R_1 + \Upsilon) &\leq V(A_1; \bar{\mathbf{s}}, R_1 + \Upsilon) \\ &< V(B; \mathbf{s}^*, R_1 + \Upsilon) \\ &= V(B; \mathbf{s}', R_1 + \Upsilon), \end{aligned}$$

where the first inequality follows from the second requirement for Υ to be a complex transfer function, and the second inequality follows from the definition of δ and $A_1 \notin X(\mathbf{s}^*; R_1 + \Upsilon)$. This contradicts $A_1 \in X(\mathbf{s}'; R_1 + \Upsilon)$.

In the second case, $A_1 \subset B$. Then $A \setminus B \subset A_2$. Since $|A \setminus B| \geq 2$, we know for each $d \in A \setminus B$, $s'_d = \hat{s}_d \leq R_2(A_2) - R_2(A_2 - d) = \bar{s}_d = s^*_d - \delta$, implying $s'_d < s^*_d$.

Hence,

$$\begin{aligned} V(A;\mathbf{s}',R_1+\Upsilon) &> V(A;\mathbf{s}^*,R_1+\Upsilon) \\ &= V(B;\mathbf{s}^*,R_1+\Upsilon) \\ &= \Pi(\mathbf{s}^*;R_1+\Upsilon) \\ &\geq V(A_1;\mathbf{s}^*,R_1+\Upsilon) \\ &= V(A_1;\mathbf{s}',R_1+\Upsilon), \end{aligned}$$

where the first inequality follows from the the second requirement for Υ to be a complex transfer function. This contradicts $A_1 \in X(\mathbf{s}';R_1+\Upsilon)$. □

OA.2.12 Proofs of Propositions 7 and 8

Proof of Proposition 7. For necessity, suppose that R satisfies the chain-constrained substitutes condition, but there exists $\mathbf{s} \in \mathbb{R}^D$, $m \in [1,M]_{\mathbb{Z}}$, and $A \in X(\mathbf{s};R,\mathcal{D}_{m-1})$ such that for all $B \in X(\mathbf{s};R,\mathcal{D}_m)$, $A \not\subset B$. Among demand sets in $X(\mathbf{s};R,\mathcal{D}_m)$, pick A^* such that for all $B \in X(\mathbf{s};R,\mathcal{D}_m)$, $|B \setminus A| \geq |A^* \setminus A|$. Find a small $\epsilon > 0$ such that

$$M\epsilon < \Pi(\mathbf{s};R,\mathcal{D}_m) - \max\{V(B;\mathbf{s},R) : B \in \mathcal{D}_m \setminus X(\mathbf{s};R,\mathcal{D}_m)\}.$$

Construct $\mathbf{s}'$ such that $s'_d = s_d$ for all $d \in A$, and $s'_d = s_d + \epsilon$ for all $d \in D \setminus A$. The construction entails $A^* \in X(\mathbf{s}';R,\mathcal{D}_m)$ and $X(\mathbf{s}';R,\mathcal{D}_{m-1}) = \{A\}$. By the chain-constrained substitutes condition, $A \subset A^*$, a contradiction.

For sufficiency, suppose that the assumption of the sufficiency claim holds for R, but the chain-constrained substitutes condition fails: there exists $\mathbf{s} \in \mathbb{R}^D$, $m \in [1,M]_{\mathbb{Z}}$, and $A \in X(\mathbf{s};R,\mathcal{D}_m)$ such that for all $B \in X(\mathbf{s};R,\mathcal{D}_{m-1})$, $B \not\subset A$. Among demand sets in $X(\mathbf{s};R,\mathcal{D}_{m-1})$, pick A^* such that for all $B \in X(\mathbf{s};R,\mathcal{D}_{m-1})$, $|B \setminus A| \geq |A^* \setminus A|$. Find a small $\epsilon > 0$ such that

$$M\epsilon < \Pi(\mathbf{s};R,\mathcal{D}_{m-1}) - \max\{V(B;\mathbf{s},R) : B \in \mathcal{D}_{m-1} \setminus X(\mathbf{s};R,\mathcal{D}_{m-1})\}.$$

Construct $\mathbf{s}'$ such that $s'_d = s_d$ for all $d \in A$, and $s'_d = s_d + \epsilon$ for all $d \in D \setminus A$. The construction entails $A^* \in X(\mathbf{s}';R,\mathcal{D}_{m-1})$ and $X(\mathbf{s}';R,\mathcal{D}_m) = \{A\}$. By the assumption of the sufficiency claim, $A^* \subset A$, a contradiction. □

Proof of Proposition 8. For necessity, consider a revenue function R satisfying the substitutes condition. First, because an exact constraint is a special type of generalized interval constraints, by Theorem 1 of KSY, R satisfies the exactly-constrained substitutes condition.

To prove the chain-constrained substitutes condition, consider $\mathbf{s} \in \mathbb{R}^D$, $m \in [1, M]_{\mathbb{Z}}$, and $A \in X(\mathbf{s}; R, \mathcal{D}_m)$. Let T be a cardinally concave transfer function such that $T(C) = -\sum_{i=0}^{||C|-m|} i$ for each $C \subset D$, and define

$$\alpha := 3 \max \left\{ \left| R(A) - \sum_{d \in A} s_d \right| : A \subset D \right\}.$$

Then a firm facing $R + \alpha T$ maximizes profit only if it hires exactly m workers. Since a cardinal transfer function maintains the relative profitability of worker sets of the same cardinality, we have $X(\mathbf{s}; R + \alpha T) = X(\mathbf{s}; R, \mathcal{D}_m)$. So $A \in X(\mathbf{s}; R + \alpha T)$, and thus by the second part of Lemma 11 in the Online Appendix of KSY, there exists $A' \in X(\mathbf{s}; R + \alpha T, \mathcal{D}_{m-1})$ such that $A' \subset A$. But again, we know $X(\mathbf{s}; R + \alpha T, \mathcal{D}_{m-1}) = X(\mathbf{s}; R, \mathcal{D}_{m-1})$, so $A' \in X(\mathbf{s}; R, \mathcal{D}_{m-1})$.

To prove concavity in scale, suppose otherwise for contradiction that R is not concave in scale. Then there exists $\mathbf{s} \in \mathbb{R}^D$ and $m \in [1, M-1]_{\mathbb{Z}}$ such that $2\Pi(\mathbf{s}; R, \mathcal{D}_m) < \Pi(\mathbf{s}; R, \mathcal{D}_{m+1}) + \Pi(\mathbf{s}; R, \mathcal{D}_{m-1})$. By the chain-constrained substitutes condition, there exists $A \subset D$ and $d, d' \in D \setminus A$ such that $A \in X(\mathbf{s}; R, \mathcal{D}_{m-1})$, $A + d \in X(\mathbf{s}; R, \mathcal{D}_m)$, and $A + d + d' \in X(\mathbf{s}; R, \mathcal{D}_{m+1})$.

Define a new revenue function R' such that $R'(B) = R(B) - \sum_{d \in B} s_d$ for each $B \subset D$. By Proposition 1, R' also satisfies the substitutes condition, and is thus discrete concave by Lemma OA.1 of the Online Appendix. Therefore,

$$\begin{aligned}
\Pi(\mathbf{s}; R, \mathcal{D}_{m+1}) + \Pi(\mathbf{s}; R, \mathcal{D}_{m-1}) &= V(A + d + d'; \mathbf{s}, R) + V(A; \mathbf{s}, R) \\
&= R'(A + d + d') + R'(A) \\
&\leq R'(A + d) + R'(A + d') \\
&= V(A + d; \mathbf{s}, R) + V(A + d'; \mathbf{s}, R) \\
&= \Pi(\mathbf{s}; R, \mathcal{D}_m) + V(A + d'; \mathbf{s}, R) \\
&\leq 2\Pi(\mathbf{s}; R, \mathcal{D}_m),
\end{aligned}$$

where the first inequality is due to the first part of Theorem OA.2 of the Online Appendix. But this contradicts $2\Pi(\mathbf{s}; R, \mathcal{D}_m) < \Pi(\mathbf{s}; R, \mathcal{D}_{m+1}) + \Pi(\mathbf{s}; R, \mathcal{D}_{m-1})$.

For sufficiency, let a revenue function R satisfy the three assumptions. By Lemma OA.6 of the Online Appendix, we only need to demonstrate the single-improvement property of R. Start with an arbitrary $\mathbf{s} \in \mathbb{R}^D$ and $A \subset D$ such that $A \notin X(\mathbf{s}; R)$. First consider the case in which $A \notin X(\mathbf{s}; R, \mathcal{D}_{|A|})$. By the exactly-constrained substitutes condition, $X(\,\cdot\,; R, \mathcal{D}_{|A|})$ satisfies the substitutes

condition and thus the single-improvement property (Lemma OA.6). Hence there exists $A' \in \mathcal{D}_{|A|}$ such that $V(A;\mathbf{s},R) < V(A';\mathbf{s},R)$, $|A \setminus A'| = 1$, and $|A' \setminus A| = 1$, as desired.

Consider the other case in which $A \in X(\mathbf{s};R,\mathcal{D}_{|A|})$. Since $A \notin X(\mathbf{s};R)$, we know $\Pi(\mathbf{s};R,\mathcal{D}_{|A|}) < \Pi(\mathbf{s};R)$. Therefore, by concavity in scale, we know that either $\Pi(\mathbf{s};R,\mathcal{D}_{|A|}) < \Pi(\mathbf{s};R,\mathcal{D}_{|A|-1})$ or $\Pi(\mathbf{s};R,\mathcal{D}_{|A|}) < \Pi(\mathbf{s};R,\mathcal{D}_{|A|+1})$. In the first subcase, the chain-constrained substitutes condition entails the existence of $A' \in X(\mathbf{s};R,\mathcal{D}_{|A|-1})$ such that $V(A;\mathbf{s},R) < V(A';\mathbf{s},R)$, $|A \setminus A'| = 1$, and $|A' \setminus A| = 0$, as desired. In the second subcase, Proposition 7 entails the existence of $A' \in X(\mathbf{s};R,\mathcal{D}_{|A|+1})$ such that $V(A;\mathbf{s},R) < V(A';\mathbf{s},R)$, $|A' \setminus A| = 1$, and $|A \setminus A'| = 0$, as desired. □

OA.2.13 Proofs of Theorem 5 and Corollary 3

Proof of Theorem 5. For sufficiency, take a revenue function R satisfying both the exactly-constrained substitutes condition and the chain-constrained substitutes condition. Construct a cardinal transfer function T such that $T(A) = \sum_{m=0}^{|A|} \alpha_m$ for each $A \subset D$, where $\alpha_0 = \alpha_1 = 0$, and α_m for each $m \in [2,M]_{\mathbb{Z}}$ is defined iteratively such that $\alpha_m - \alpha_{m-1}$ equals

$$-\max\{|R(B)+R(B-d-d')-R(B-d)-R(B-d')| : B \in \mathcal{D}_m,\ d,d' \in B, \text{ and } d \neq d'\}.$$

Since α_m is weakly decreasing, T is cardinally concave. We only need to show that $R+T$ satisfies the substitutes condition.

Equivalently, we only need to show that for all $\mathbf{s} \in \mathbb{R}^D$, $A \in X(\mathbf{s};R+T)$, $d \in A$, and $\delta \in \mathbb{R}$ with $\delta > 0$, there exists $A' \in X(\mathbf{s}+\delta\mathbf{e}^d;R+T)$ satisfying $A - d \subset A'$. A simple case occurs when $\Pi(\mathbf{s}+\delta\mathbf{e}^d;R+T) = \Pi(\mathbf{s};R+T) - \delta$. In such a case, $A \in X(\mathbf{s}+\delta\mathbf{e}^d;R+T)$. By setting $A' = A$, we are done.

We are left with the other case of $\Pi(\mathbf{s}+\delta\mathbf{e}^d;R+T) > \Pi(\mathbf{s};R+T) - \delta$. In such a case, for all $B \in X(\mathbf{s}+\delta\mathbf{e}^d;R+T)$, $d \notin B$. Set $\delta' \coloneqq \Pi(\mathbf{s};R+T) - \Pi(\mathbf{s}+\delta\mathbf{e}^d;R+T)$ and $\mathbf{s}' \coloneqq \mathbf{s} + \delta'\mathbf{e}^d$. This implies $\delta' \in [0,\delta)$. Also, for all $B \in X(\mathbf{s}+\delta\mathbf{e}^d;R+T)$ and

$C \subset D$ with $d \notin C$,

$$\begin{aligned} V(A; \mathbf{s}', R+T) &= \Pi(\mathbf{s}; R+T) - \delta' \\ &= \Pi(\mathbf{s} + \delta \mathbf{e}^d; R+T) \\ &= V(B; \mathbf{s} + \delta \mathbf{e}^d, R+T) \\ &= V(B; \mathbf{s}', R+T) \\ &\geq V(C; \mathbf{s}', R+T). \end{aligned}$$

Further, $A \in X(\mathbf{s}; R+T)$ implies $V(A; \mathbf{s}', R+T) \geq V(C; \mathbf{s}', R+T)$ for any $C \subset D$ with $d \in C$. In total, $V(A; \mathbf{s}', R+T) \geq V(C; \mathbf{s}', R+T)$ for any $C \subset D$, i.e., $A \in X(\mathbf{s}'; R+T)$. In addition, for all $B \in X(\mathbf{s} + \delta \mathbf{e}^d; R+T)$, we have shown $V(B; \mathbf{s}', R+T) = V(A; \mathbf{s}', R+T)$, and thus $B \in X(\mathbf{s}'; R+T)$, i.e., $X(\mathbf{s} + \delta \mathbf{e}^d; R+T) \subset X(\mathbf{s}'; R+T)$.

Consider a subcase in which there exists $B^* \in X(\mathbf{s}'; R+T)$ such that $d \notin B^*$ and $|B^*| \geq |A|$. By iteratively applying the chain-constrained substitutes condition, we can find $B' \subset B^*$ such that $B' \in X(\mathbf{s}'; R+T, \mathcal{D}_{|A|})$. Since $A \in X(\mathbf{s}'; R+T)$, we also have $X(\mathbf{s}'; R+T, \mathcal{D}_{|A|}) \subset X(\mathbf{s}'; R+T)$. Thus $B' \in X(\mathbf{s}'; R+T)$. Since $d \notin B'$, $B' \in X(\mathbf{s} + \delta \mathbf{e}^d; R+T)$. Because $|B'| = |A|$, we further have $\Pi(\mathbf{s} + \delta \mathbf{e}^d; R+T, \mathcal{D}_{|A|}) = \Pi(\mathbf{s} + \delta \mathbf{e}^d; R+T)$.

Since a cardinal transfer function maintains the relative profitability of worker sets of the same cardinality, $A \in X(\mathbf{s}'; R+T)$ implies $A \in X(\mathbf{s}'; R, \mathcal{D}_{|A|})$. Given $\mathbf{s} + \delta \mathbf{e}^d \geq \mathbf{s}'$, we can apply Definition 4, and establish the existence of $A' \in X(\mathbf{s} + \delta \mathbf{e}^d; R, \mathcal{D}_{|A|})$ satisfying $A - d = \{d' \in A : s_{d'} = s'_{d'}\} \subset A'$. We are done because

$$A' \in X(\mathbf{s} + \delta \mathbf{e}^d; R, \mathcal{D}_{|A|}) = X(\mathbf{s} + \delta \mathbf{e}^d; R+T, \mathcal{D}_{|A|}) \subset X(\mathbf{s} + \delta \mathbf{e}^d; R+T),$$

where the inclusion follows from the conclusion of the last paragraph.

In the other subcase, every $B \in X(\mathbf{s}'; R+T)$ with $d \notin B$ has $|B| < |A|$. Then due to $A \in X(\mathbf{s}'; R+T)$, we have $d \in B$ for all $B \in X(\mathbf{s}'; R+T, \mathcal{D}_{|A|})$.

We first show that there exists $B^* \in X(\mathbf{s}'; R+T)$ such that $d \notin B^*$ and $|B^*| = |A| - 1$ by contradiction. Suppose otherwise. Then for all $B \in X(\mathbf{s}'; R+T)$, $d \notin B$ implies $|B| < |A| - 1$. Since $X(\mathbf{s} + \delta \mathbf{e}^d; R+T) \subset X(\mathbf{s}'; R+T)$ and for all $B \in X(\mathbf{s} + \delta \mathbf{e}^d; R+T)$ we have $d \notin B$, there exists $B^\circ \in X(\mathbf{s}'; R+T)$ with $d \notin B^\circ$. Pick the above B° such that $|B| \leq |B^\circ| < |A| - 1$ for all $B \in X(\mathbf{s}'; R+T)$ with $d \notin B$.

According to Proposition 9, $R+T$ satisfies both the exactly-constrained substitutes condition and the chain-constrained substitutes condition. By iteratively applying Definition 5 to $A \in X(\mathbf{s}'; R+T, \mathcal{D}_{|A|})$, there exists $B' \in X(\mathbf{s}'; R+T, \mathcal{D}_{|B^\circ|})$ satisfying $B' \subset A$. Because $B^\circ \in X(\mathbf{s}'; R+T)$,

$$B' \in X(\mathbf{s}'; R+T, \mathcal{D}_{|B^\circ|}) \subset X(\mathbf{s}'; R+T).$$

Denote $A \setminus B'$ by $\{d_1, \cdots, d_{|A|-|B^\circ|}\}$ such that $d_1 \neq d$. Then

$$\begin{aligned} & V(A; \mathbf{s}', R+T) - V(A-d_1; \mathbf{s}', R+T) \\ = \; & R(A) - R(A-d_1) + \alpha_{|A|} - \mathbf{s}(d_1) \\ \leq \; & R(A-d_2) - R(A-d_1-d_2) + \alpha_{|A|-1} - \mathbf{s}(d_1), \end{aligned}$$

where the inequality follows from the definition of transfer function T. Further iteratively applying the logic behind the inequality above (if necessary), we have

$$\begin{aligned} & V(A; \mathbf{s}', R+T) - V(A-d_1; \mathbf{s}', R+T) \\ \leq \; & R(A-d_2-\ldots-d_{|A|-|B^\circ|}) - R(A-d_1-\ldots-d_{|A|-|B^\circ|}) + \alpha_{|B^\circ|+1} - \mathbf{s}(d_1) \\ = \; & R(B'+d_1) - R(B') + \alpha_{|B^\circ|+1} - \mathbf{s}(d_1) \\ = \; & V(B'+d_1; \mathbf{s}', R+T) - V(B'; \mathbf{s}', R+T). \end{aligned}$$

It follows that

$$\begin{aligned} V(B'+d_1; \mathbf{s}', R+T) &\geq V(A; \mathbf{s}', R+T) + V(B'; \mathbf{s}', R+T) - V(A-d_1; \mathbf{s}', R+T) \\ &= 2\Pi(\mathbf{s}'; R+T) - V(A-d_1; \mathbf{s}', R+T) \\ &\geq \Pi(\mathbf{s}'; R+T). \end{aligned}$$

Thus, $B'+d_1 \in X(\mathbf{s}'; R+T)$ while $d \notin (B'+d_1)$. This contradicts the definition of B°.

Take $B^* \in X(\mathbf{s}'; R+T)$ satisfying $d \notin B^*$ and $|B^*| = |A|-1$. For any $\epsilon > 0$, we have $B^* \in X(\mathbf{s}'+\epsilon\mathbf{e}^d; R+T)$, and thus $B^* \in X(\mathbf{s}'+\epsilon\mathbf{e}^d; R+T, \mathcal{D}_{|A|-1})$. Then $d \notin B'$ for all $B' \in X(\mathbf{s}'+\epsilon\mathbf{e}^d; R+T, \mathcal{D}_{|A|-1})$. Now recall that $d \in B$ for all $B \in X(\mathbf{s}'; R+T, \mathcal{D}_{|A|})$, so we can pick a sufficiently small ϵ such that $A \in X(\mathbf{s}'+\epsilon\mathbf{e}^d; R+T, \mathcal{D}_{|A|})$. By Definitions 5, there exists $A' \subset A$ satisfying $A' \in X(\mathbf{s}'+\epsilon\mathbf{e}^d; R+T, \mathcal{D}_{|A|-1})$. Since $d \notin A'$, we have $A' = A-d$. Since $B^* \in X(\mathbf{s}'; R+T)$, we have $A' \in X(\mathbf{s}'; R+T)$. It follows that $A' \in X(\mathbf{s}+\delta\mathbf{e}^d; R+T)$, and we are done.

For necessity, assume that the sum of a revenue function R and a cardinally

concave transfer function T satisfies the substitutes condition. By Theorem 1 of KSY, for any $m \in [1, M]_{\mathbb{Z}}$, $X(\,\cdot\,; R+T, \mathcal{D}_m)$ also satisfies the substitutes condition. Because a cardinal transfer function maintains the relative profitability of worker sets of the same cardinality, we have $X(\cdot; R+T, \mathcal{D}_m) = X(\cdot; R, \mathcal{D}_m)$. Hence $X(\cdot; R, \mathcal{D}_m)$ satisfies the substitutes condition, i.e., R satisfies the exactly-constrained substitutes condition.

To establish the chain-constrained substitutes condition, start with any $\mathbf{s} \in \mathbb{R}^D$, $m \in [1, M]_{\mathbb{Z}}$, and $A \in X(\mathbf{s}; R, \mathcal{D}_m)$. Again, because a cardinal transfer function maintains the relative profitability of worker sets of the same cardinality, we have $A \in X(\mathbf{s}; R+T, \mathcal{D}_m)$. Since $R+T$ satisfies the substitutes condition, by Proposition 8, there exists $A' \in X(\mathbf{s}; R+T, \mathcal{D}_{m-1})$ such that $A' \subset A$. But we also have $A' \in X(\mathbf{s}; R, \mathcal{D}_{m-1})$. □

Proof of Corollary 3. Let $\mathcal{R}$ be a bounded class of revenue functions, each satisfying the exactly-constrained substitutes condition and the chain-constrained substitutes condition. Following the construction in the proof of Theorem 5, for each $R \in \mathcal{R}$, we can find T^R such that $T^R(A) = \sum_{m=0}^{|A|} \alpha_m^R$ for each $A \subset D$, where $\alpha_0^R = \alpha_1^R = 0$, and α_m^R for each $m \in [2, M]_{\mathbb{Z}}$ is defined iteratively such that $\alpha_m^R - \alpha_{m-1}^R$ equals

$$-\max\{|R(B)+R(B-d-d')-R(B-d)-R(B-d')| : B \in \mathcal{D}_m,\ d, d' \in B, \text{ and } d \neq d'\}.$$

We know from the proof of Theorem 5 that for all $R \in \mathcal{R}$, $R + T^R$ satisfies the substitutes condition.

Construct a cardinal transfer function T such that $T(A) = \sum_{m=0}^{|A|} \alpha_m$ for each $A \subset D$, where $\alpha_0 = \alpha_1 = 0$, and α_m for each $m \in [2, M]_{\mathbb{Z}}$ is defined iteratively such that $\alpha_m - \alpha_{m-1} = \inf\{\alpha_m^R - \alpha_{m-1}^R : R \in \mathcal{R}\}$. Since $\mathcal{R}$ is bounded, $\alpha_m - \alpha_{m-1} \leq 0$ is also bounded for each $m \in [2, M]_{\mathbb{Z}}$, and thus T is well-defined. For all $R \in \mathcal{R}$, we have $\alpha_m - \alpha_{m-1} \leq \alpha_m^R - \alpha_{m-1}^R$, so T is more concave than T^R. According to Proposition 6, T reestablishes the substitutes condition for all $R \in \mathcal{R}$.

OA.2.14 Proof of Corollary 5

Proof of Corollary 5. For sufficiency, start with a trade transfer function $v = v_1 + v_2$, where v_1 and v_2 respectively derive from an additively separable transfer function T_1 and a cardinally concave transfer function T_2. Let u be an arbitrary fully substitutable trade revenue function which derives from R^u. By Theorem 9, R^u satisfies the substitutes condition, so, by the sufficiency part of Theorem 1, $R^u + T_1 + T_2$ satisfies the substitutes condition. But $u + v = u + v_1 + v_2$ derives from

$R^u + T_1 + T_2$. Theorem 9 tells us that $u + v$ is fully substitutable. Thus v always preserves full substitutability.

For necessity, let us prove the contrapositive. Start with a trade transfer function v which derives from a transfer function T^v but cannot be written as the sum of a derived additively separable trade transfer function and a derived cardinally concave one. Then it is simple to prove by contradiction that T^v cannot be written as the sum of an additively separable transfer function and a cardinally concave one. According to the necessity part of Theorem 1, there exists a revenue function R satisfying the substitutes condition such that $R + T^v$ fails the condition. Let u^R be the trade revenue function deriving from R. By Theorem 9, u^R is fully substitutable. Also, $u^R + v$ derives from $R + T^v$, so, by Theorem 9, is not fully substitutable. Thus v fails to always preserve full substitutability. □